

Pathogen profile

***Botrytis cinerea*: the cause of grey mould disease**BRIAN WILLIAMSON¹, BETTINA TUDZYNSKI², PAUL TUDZYNSKI² AND JAN A. L. VAN KAN^{3,*}¹Scottish Crop Research Institute, Invergowrie, Dundee DD2 5DA, UK²Westfälische Wilhelms-Universität Münster, Institute of Botany, Schlossgarten 3, 48149 Münster, Germany³Wageningen University, Laboratory of Phytopathology, Binnenhaven 5, 6709 PD Wageningen, the Netherlands**SUMMARY**

Introduction: *Botrytis cinerea* (teleomorph: *Botryotinia fuckeliana*) is an airborne plant pathogen with a necrotrophic lifestyle attacking over 200 crop hosts worldwide. Although there are fungicides for its control, many classes of fungicides have failed due to its genetic plasticity. It has become an important model for molecular study of necrotrophic fungi.

Taxonomy: Kingdom: Fungi, phylum: Ascomycota, subphylum: Pezizomycotina, class: Leotiomycetes, order: Helotiales, family: Sclerotiniaceae, genus: *Botryotinia*.

Host range and symptoms: Over 200 mainly dicotyledonous plant species, including important protein, oil, fibre and horticultural crops, are affected in temperate and subtropical regions. It can cause soft rotting of all aerial plant parts, and rotting of vegetables, fruits and flowers post-harvest to produce prolific grey conidiophores and (macro)conidia typical of the disease.

Pathogenicity: *B. cinerea* produces a range of cell-wall-degrading enzymes, toxins and other low-molecular-weight compounds such as oxalic acid. New evidence suggests that the pathogen triggers the host to induce programmed cell death as an attack strategy.

Resistance: There are few examples of robust genetic host resistance, but recent work has identified quantitative trait loci in tomato that offer new approaches for stable polygenic resistance in future.

Useful websites: <http://www.phi-base.org/query.php>, http://www.broad.mit.edu/annotation/genome/botrytis_cinerea/Home.html, <http://urgi.versailles.inra.fr/projects/Botrytis/>, <http://cogeme.ex.ac.uk>

INTRODUCTION

The pathogen *Botrytis cinerea* Pers. Fr. (teleomorph *Botryotinia fuckeliana* (de Bary) Whetzel) causes serious losses in more than

200 crop species worldwide. It is most destructive on mature or senescent tissues of dicotyledonous hosts, but it usually gains entry to such tissues at a much earlier stage in crop development and remains quiescent for a considerable period before rapidly rotting tissues when the environment is conducive and the host physiology changes. Therefore, serious damage is caused following harvest of apparently healthy crops and the subsequent transport to distant markets where the losses become evident. However, *B. cinerea* also causes massive losses in some field- and greenhouse-grown horticultural crops prior to harvest, or even at the seedling stage in some hosts. Some monocotyledonous hosts are also susceptible to attack by *B. cinerea*, and there is a group of related *Botrytis* species specialized to infect about a dozen monocot hosts. The latter group will not be discussed in this review, but their evolution and biology pose interesting questions about the nature of host specialization and the dynamics of speciation within the genus (Staats *et al.*, 2005, 2007).

B. cinerea is difficult to control because it has a variety of modes of attack, diverse hosts as inoculum sources, and it can survive as mycelia and/or conidia or for extended periods as sclerotia in crop debris. For these reasons the use of any single control measure is unlikely to succeed and a more detailed understanding of the host–pathogen interaction, the microenvironment in which the fungus operates and its microbial competitors on the host is essential. The current cost of bringing a new fungicide or biological control agent to market is so high that only major crops attract sufficient interest by agribusiness.

A major textbook on *Botrytis* species and the diseases they cause was published (Elad *et al.*, 2004) which up-dated the classical text on this subject by Coley-Smith *et al.* (1980). This profile will concentrate on the advances made in the understanding of the molecular interplay between the pathogen and certain key hosts gained especially in the last 5 years. Because of the worldwide importance of this fungus and the availability of molecular genetic tools to study the organism, it has become the most extensively studied necrotrophic fungal pathogen. Its entire genome sequence will be available for analysis in the near future and this should provide new insight into the biology, evolution and opportunities for control of this organism.

* Correspondence: E-mail: jan.vankan@wur.nl

TAXONOMY AND GENETIC VARIATION

Phylogenetic analysis of 22 species of the genus *Botrytis* revealed that *B. cinerea* forms a small clade with three other species, all of which are specialized pathogens of dicots (Staats *et al.*, 2005). Beever and Weeds (2004) reviewed the current status of *B. cinerea* and its teleomorph *Botryotinia fuckeliana*. Consequently, only a brief outline will be given here. The conidia (macroconidia) are multinucleate and the microconidia (male spermatia) are uninucleate. Shirane *et al.* (1988) reported 16 chromosomes at mitotic metaphase and Faretra and Grindle (1992) also found 16 chromosomes in developing asci. Apothecia of *B. cinerea* are rare in the field, but are more common in other *Botrytis* spp. Most strains are heterothallic, carrying one or other allele of the mating type locus *MAT1-1* or *MAT1-2* (Faretra *et al.*, 1988); however, there are also field isolates with dual mating phenotypes (Faretra and Pollastro, 1996; van der Vlugt-Bergmans *et al.*, 1993). Sexual crossing *in vitro* involves incubating sclerotia of the female (sclerotial) parent for long periods at zero degrees (preconditioning) before fertilizing them with a suspension of microconidia from the spermatial parent obtained by irrigation of an ageing culture (Faretra *et al.*, 1988).

Genetic variation in *B. cinerea* populations has been studied using a variety of molecular techniques including RFLP analysis of PCR-amplified loci (Giraud *et al.*, 1997), PCR detection of transposable elements (Diolez *et al.*, 1995; Levis *et al.*, 1997a), RAPD fingerprinting (Kerssies *et al.*, 1997; van der Vlugt-Bergmans *et al.*, 1993), amplified fragment length polymorphism (AFLP) analysis (Moyano *et al.*, 2003), fingerprinting of repetitive sequences by microsatellite primed (MP)-PCR (Ma and Michailides, 2005), PCR amplification of microsatellite loci (Fournier *et al.*, 2002) and DNA sequencing of gene regions (Albertini and Leroux, 2004; Albertini *et al.*, 2002; Fournier *et al.*, 2003). Based on multiple gene genealogies, Fournier *et al.* (2005) postulated that *B. cinerea* is a species complex comprising two phylogenetic species, but this hypothesis has not yet been fully adopted by the *Botrytis* community.

HOST RANGE

Droby and Lichter (2004) provide a comprehensive list of post-harvest rots caused by *B. cinerea*; these range from grey mould on different plant organs, including flowers, fruits, leaves, shoots and soil storage organs (i.e. carrot, sweet potato), although the fungus is not regarded as a true root pathogen or one causing soil-borne diseases. Vegetables (i.e. cabbage, lettuce, broccoli, beans) and small fruit crops (grape, strawberry, raspberry, blackberry) are most severely affected. With increasing international trade in cold-stored produce this fungus has attained great importance because it can grow effectively over long periods at just above freezing temperatures in products such as kiwifruit,

apples ('dry eye rot') and pears. Similarly, the important trade in cut flowers is adversely affected by this pathogen; rose and gerbera flowers are particularly prone to damage. Culture of plants out-of-season in heated or unheated greenhouses and under plastic tunnels used increasingly to supply fruits, vegetables, herbs and flowers in northern latitudes greatly increases the risk of infection, especially in tomato, cucumber and sweet pepper.

There are important field crops that sustain serious damage due to grey mould. Most notable are the heavy losses in chick-peas and other protein-rich legumes that support millions of rural families in India, Bangladesh and Nepal (Pande *et al.*, 2002). French bean (*Phaseolus vulgaris*) can sustain almost complete losses. Most legumes suffer attack by *B. cinerea* to some extent, but field bean (*Vicia faba*) is also damaged by chocolate spot caused by *B. fabae*. Blossom blight in alfalfa in Canada causes serious problems in seed production in irrigated areas (Gossen and Platford, 1999). Sunflower is an important oil crop that can be infected severely. In tree nurseries, conifer seedlings are vulnerable to grey mould. With increasing interest in renewable technologies and sustainable development it is important to note also that industrial hemp (*Cannabis sativa*) used for fibre production is susceptible to head blight.

SYMPTOMS

B. cinerea is responsible for a very wide range of symptoms (Fig. 1) and these cannot easily be generalized across plant organs and tissues. Soft rots, accompanied by collapse and water-soaking of parenchyma tissues, followed by a rapid appearance of grey masses of conidia are perhaps the most typical symptoms on leaves and soft fruits (Fig. 1a,b). In thick-skinned fruits, such as kiwifruits, the dark water-soaking symptom is evident only after cutting. On many fruits and vegetables the infection commonly begins on attached senescent flowers and then as a soft rot it spreads to affect the adjacent developing fruit (blossom-end rot), as in courgettes (zucchini), cucumbers, French beans, strawberries and apples. On flower petals, symptoms range from minute 'pock' marks to full-scale soft rotting (Fig. 1c) depending on the environmental conditions. In greenhouse-grown tomato, the greatest damage occurs on stems at pruning wounds where the fungus can rot through the entire stem. Soft rotting of mature tomato fruits occurs mainly post-harvest; an unusual 'ghost spot' symptom in unripe tomato is associated with a successful host defence, but the symptom renders fruits unmarketable.

In red raspberry (*Rubus idaeus*), apart from the devastating grey mould on fruits, the pathogen attacks mature-to-senescent leaves causing a wedge-shaped chestnut brown lesion with yellow margin that spreads to the node on vegetative stems (primocanes) to give rise to a conspicuous pale brown fast-spreading lesion (up to 15 cm) in the primary cortex of the stem (Fig. 1d). Such infection does not enter the axillary buds because

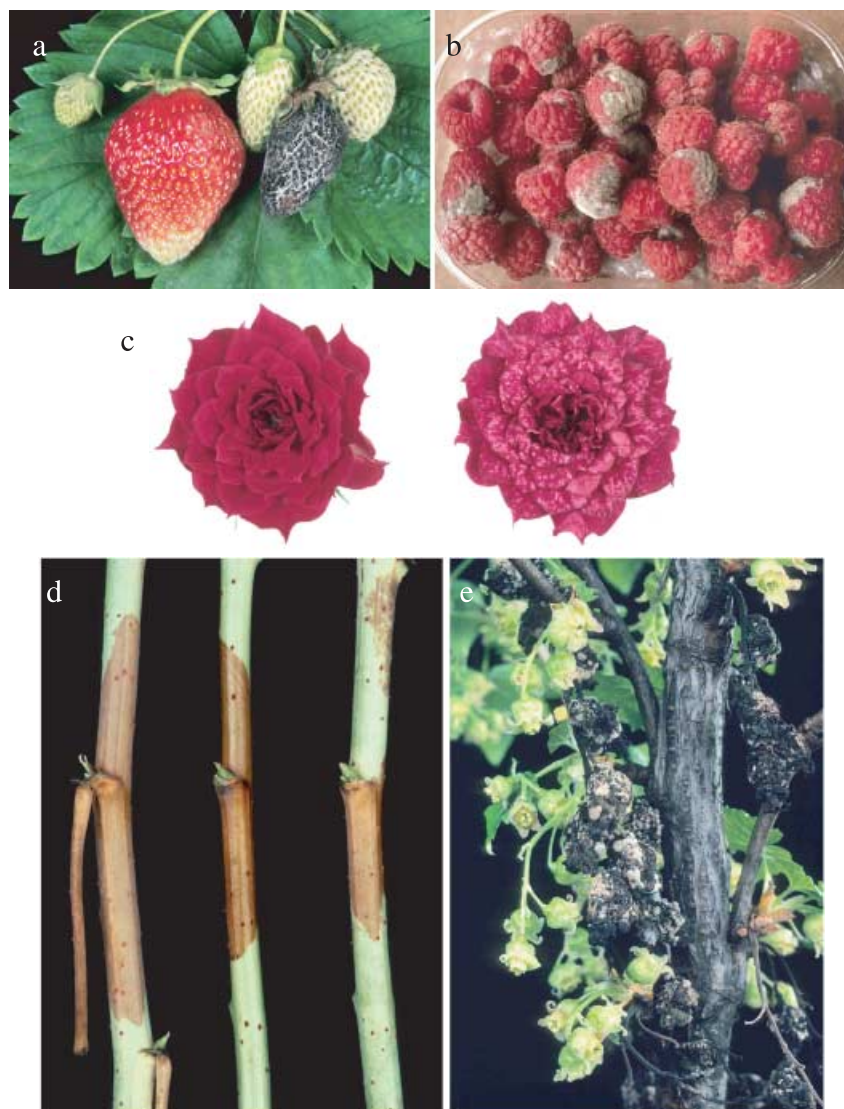


Fig. 1 Symptoms of infection by *Botrytis cinerea*. (a) Grey mould of strawberry fruit. (b) Grey mould of raspberry fruit. (c) Rose petals blemished by lesions (right) following inoculation with dry conidia and incubation at 100% RH for 48 h, compared with non-inoculated control (left). Reprinted from Williamson *et al.* (1995). Effect of humidity on infection of rose petals by dry-inoculated conidia of *Botrytis cinerea*. *Mycological Research* **99**: 1303–1310, with permission from Elsevier. (d) Lesions arising at nodes following infection of raspberry leaves in autumn. (e) Mummified 1-year-old blackcurrant fruits attached to stem, releasing conidia amongst newly opened flowers.

of periderm layers, but it retards development of the buds at infected nodes with the result that they fail to produce fertile lateral shoots in the following year. After winter dormancy, the stem lesions in raspberry become white and show large black sclerotia that produce masses of grey conidia in spring. In blackcurrant, symptomless infection of flower styles (detected by fluorescence microscopy) by *B. cinerea* leads to premature abscission of developing fruits associated with ethylene generation in a condition called 'run-off'.

Seed-borne infection has been reported in over 50 hosts, including flax, sunflower and lettuce (see Maude, 1980). Seed transmission occurs in chickpeas, and in Australia it can cause total crop failure (Burgess *et al.*, 1997). Grey mould in this crop often begins by rotting of the herbaceous stems at ground level, with other soft-rot lesions appearing also on leaves and pods.

Barnes and Shaw (2003) described the occurrence of widespread internal infection in *Primula* × *polyantha* plants grown from infected commercial seeds with symptoms of disease only appearing 3 months later at flowering. This apparent endophytic relationship with the host remains to be studied by modern microscopic tools.

LIFE CYCLE AND EPIDEMIOLOGY

Sclerotia develop within dying host tissues and represent an important survival mechanism in *B. cinerea*, but they are very variable in size, and are not readily apparent in all susceptible crops. The melanized rind and β -glucans encasing the internal mycelium protect sclerotia from desiccation, UV radiation and microbial attack over long periods (Backhouse and Willets, 1984).

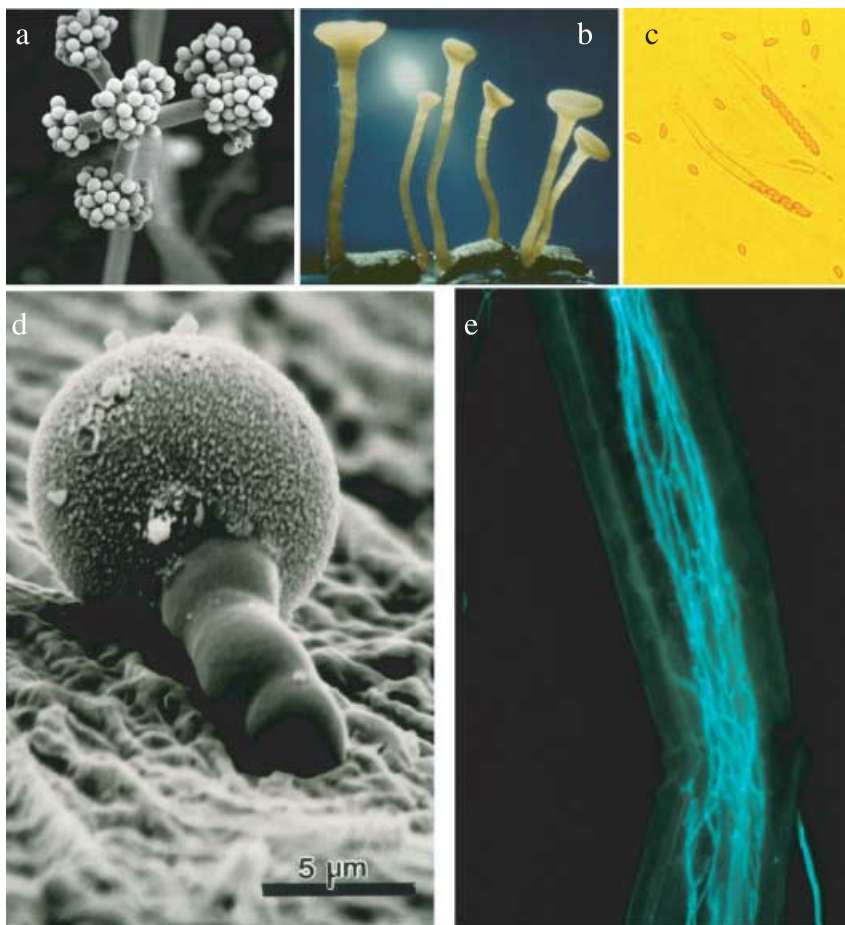


Fig. 2 (a) *B. cinerea* conidiophore with mature conidia *in situ* (LTSEM). Reproduced from the front cover of *Botrytis: Biology, Pathology and Control* (Elad, Y., Williamson, B., Tudzynski, P. and Delen, N., eds) with kind permission of Springer Science and Business Media. (b) Apothecia of *Botryotinia fuckeliana*, approximately 10 weeks after spermatization. (c) Two asci each containing eight ascospores, surrounded by ascospores released from damaged asci. (d) Conidium germinating in absence of water droplet on abaxial surface of rose petal (LTSEM reprinted from Williamson *et al.* (1995). Effect of humidity on infection of rose petals by dry-inoculated conidia of *Botrytis cinerea*. *Mycological Research* **99**: 1303–1310, with permission from Elsevier. (e) Quiescent infection in raspberry flower: hyphae of *B. cinerea* growing inside the transmitting tissues in absence of pollen tubes (aniline blue stained and viewed by fluorescence microscopy).

Sclerotia commence growth in early spring in temperate regions to produce conidiophores and multinucleate conidia (Fig. 2a), serving as a primary source of inoculum within a crop. Mycelium also survives within infected dead host tissues left as crop debris and inside some seeds to serve as primary inoculum. In perennial crops, the dead leaves, flowers and mummified fruits contain masses of mycelium that can often be ideally situated within a crop canopy to produce conidia and initiate infections. The pathogen also forms microconidia from phialides abundantly in ageing cultures, which function primarily as spermatia. The sexual cycle involves the spermatization of sclerotia, leading to the production of apothecia (Fig. 2b) and asci with eight binucleate ascospores (Fig. 2c). The cellular details of plasmogamy and initiation of apothecia have still not been described. Furthermore, the apothecia are unrecorded or rare in most crops attacked by *B. cinerea* and any conclusions about the role of the sexual cycle in the species are based mainly on molecular analysis of genetic variation (Beever and Weeds, 2004).

Conidia generated at the sources of primary inoculum follow a well-defined diurnal cycle of initiation, production and dissemination that is regulated by fluctuations in temperature and

humidity; a rapid decline in humidity with rise in temperature in early morning causes twisting and drying of conidiophores to eject conidia into air currents (Jarvis, 1962a), either individually or in small clumps (Harrison and Lowe, 1987). Water droplets can also disperse conidia, but this is probably not a major dispersal method (Jarvis, 1962b). Conidia formation is stimulated by specific wavelengths of light (Epton and Richmond, 1980) and near UV is now generally used to induce sporulation in culture. However, some isolates can sporulate in darkness. Conidia can move on air currents from neighbouring crops, yet most conidia are probably generated from primary sources within the crop. As in many fungi, the conidia contain a self-inhibitor and need to be washed to induce high germination rates *in vitro*.

B. cinerea shows remarkable flexibility in its use of different environments to germinate and obtain nutrients from a host plant. Until the 1980s, most studies on germination and initial penetration used conidial suspensions, but dry inoculations were subsequently examined in detail by Salinas *et al.* (1989), Williamson *et al.* (1987, 1995), Cole *et al.* (1996) and Coertze *et al.* (2001). It was discovered that dry-inoculated conidia produced one or two (sometimes up to five) short germ tubes and no obvious terminal

appressoria before effecting entry to an intact host cuticle on leaves, petals or fruits (Fig. 2d). An extracellular matrix was detected around only the penetration area of the germ tube following dry inoculation of bean leaves and incubation at high humidity (Cole *et al.*, 1996), whereas with conidial suspensions much longer germ tubes and extensive secretion of an extracellular matrix was observed. *B. cinerea* is able to form appressoria (Tenberge, 2004), but they are distinct from the classical types found in *Colletotrichum* or *Magnaporthe*. *B. cinerea* germlings do contain melanin in the extracellular matrix which is loosely associated with the fungal cell wall (Doss *et al.*, 2003) but they do not contain a wall that seals the appressorium from the germ tube, as would be required to enable generating extremely high osmotic pressures. It is therefore not feasible for *B. cinerea* appressoria to penetrate host tissue by physical pressure alone. If the fungus is growing strongly from a 'saprophytic base' (dead adhering petal, 'bunch trash' in grapes, pollen grains) it can form dome-shaped infection cushions on the host (Backhouse and Willets, 1987; Jarvis, 1980).

In small fruits, the floral organs are important sites for primary infection, but then the pathogen may remain inactive for a considerable period before rapidly destroying tissues at fruit maturity. In grapes there is strong histological evidence (Viret *et al.*, 2004) that conidia infect mainly the flower receptacle, and to a lesser extent the stigma and styles, and the pathogen is then held in a quiescent state by host defences. Microscopic cracks in the grape cuticle also play a part in later infections, especially if berries are swollen after rain.

The stigmatic fluid in flowers of the 'wet stigma' type serves as a nutrient medium for airborne conidia. For example, in raspberry and strawberry conidia germinate and hyphae grow slowly within the transmitting tissues of styles, following the pathway to the ovules used by pollen tubes (Fig. 2e) where the fungus survives for up to 4 weeks as a saprophyte until the fruit ripens (McNicol *et al.*, 1985). In the field, dry-inoculation of raspberry flowers with conidia resulted in a halving of the shelf life of apparently healthy fruits developed from them, compared with non-inoculated controls (Williamson *et al.*, 1987). Petals are important infection sites in many crops, and infected wind-blown petals containing mycelium can be regarded as dispersal propagules in some cases, or important sites of secondary inoculum production, as in *Phaseolus vulgaris* (Johnson and Powelson, 1983). Relative humidity (RH) is a crucial environmental factor for *B. cinerea*, but RH is extremely difficult to regulate experimentally (Harrison *et al.*, 1994). RH values above 93% are needed for conidial germination and infection of rose petals in the absence of water droplets (Williamson *et al.*, 1995). Persistence of high RH during blossom periods leads to successive cycles of infection and sporulation, leaving no opportunity for timely removal of ripening fruits and damaging epidemics can result without adequate control measures.

The role of insect vectors for *B. cinerea* has been recognized only in the last 20 years. In grapes there are several insects known to disperse viable conidia, either on their external appendages or even inside the gut, to deposit inoculum on the surface of fruits (Engelbrecht, 2002; Fermaud and Gaunt, 1995; Fermaud and Le Menn, 1989; Louis *et al.*, 1996). Although *B. cinerea* is not regarded primarily as a wound pathogen, it can infect wounds and where insects cause damage it can flourish, as in raspberry fruits infested by larvae of raspberry beetle (*Byturus tomentosus*) (Woodford *et al.*, 2002).

MOLECULAR APPROACHES FOR ANALYSIS OF GENE FUNCTION

The first successful transformation of a *Botrytis* strain was achieved by Huang *et al.* (1989) in *B. squamosa*, but it took several years before the molecular genetics of *Botrytis* was approached on a broad scale. Today, more than a dozen teams are working intensively on molecular genetics of *Botrytis* spp. and relevant molecular tools such as transformation protocols, vectors, mutants, and genomic and cDNA libraries are available. *B. cinerea* has become one of the model systems in molecular phytopathology, also stimulated by the economic damage inflicted and the resulting strong industrial interest. Indeed, the first released genome sequence (strain B05.10, 4× coverage, Broad Institute http://www.broad.mit.edu/annotation/genome/botrytis_cinerea/Home.html) was determined by an agribusiness company (Syngenta). Due to the growing number of groups interested in basic research on *B. cinerea*, a non-industrial genome sequencing initiative was started and guided by an international consortium; the sequence will soon be released (strain T4, 10× coverage, INRA/Genoscope: <http://urgi.versailles.inra.fr/projects/Botrytis/>). Apart from the great perspectives for comparative genomics, transcriptomics, proteomics and evolutionary analyses, the availability of the genome sequence has already significantly stimulated basic research in *B. cinerea*, because the identification and cloning of genes is no longer a problem. For functional analyses it is also important to estimate the genetic complexity, e.g. how many genes encoding a specific type of enzyme are present?

The high efficiency of targeted gene inactivation allows a rapid functional analysis of putative pathogenicity-related genes. The number of functionally analysed genes is rapidly expanding and even an update of the list presented in Tudzynski and Siewers (2004) will soon become obsolete. However, updated information about gene function analysis in *B. cinerea* will be accessible from a recently established pathogen–host interactions database, PHlbase (<http://www.phi-base.org/query.php>; Baldwin *et al.*, 2006). In combination with biochemical and cytological methods, these molecular genetic techniques have led to a breakthrough in our understanding of the complex biology of *B. cinerea* (van Kan, 2006).

Transformation/gene inactivation

Several selection systems are available including resistance cassettes for phleomycin, hygromycin, nourseothricin and glufosinate, allowing the generation of multiple knock-out mutants. Reporter-gene technology is still relatively underdeveloped; there are only a few reports on the successful use of β -glucuronidase (van Kan *et al.*, 1997) or GFP constructs (e.g. Rolland *et al.*, 2003; J. Schumacher and B. Tudzynski, unpublished data). A feature of *B. cinerea* transformation is the relatively low efficiency of homologous recombination when using circular vector DNA. Gene inactivation by single cross-over is difficult, whereas gene-replacement approaches using linear transformation cassettes yield high knock-out rates (70–100%). The standard technique of targeted gene inactivation has become gene replacement using long (500–1000 bp) flanks; in most laboratories these constructs are designed by PCR.

For complementation studies or integration of reporter gene constructs requiring ectopic integration, the nitrate reductase system was used successfully. Effective homologous recombination in *B. cinerea* at the *niaD* locus was first reported by Levis *et al.* (1997b). More recently, complementation and reporter gene constructs were targeted to the *niaD* locus and homologous integration was monitored by PCR or by chlorate resistance of the transformants (J. Schumacher and B. Tudzynski, unpublished data).

Good transformation rates can be obtained with the *Agrobacterium*-T-DNA transfer system (Rolland *et al.*, 2003; N. Segmüller and P. Tudzynski, unpublished data), though this system has not yet been used in *B. cinerea* for targeted gene inactivation, only for insertional mutagenesis.

As an alternative for knock-out approaches, RNAi has been used in other phytopathogens for the silencing of genes (e.g. Fitzgerald *et al.*, 2004) and it was applied successfully also in *B. cinerea* (R. Patel *et al.*, unpublished data; J. Schumacher and B. Tudzynski, unpublished data). This knock-down approach will be especially valuable for the analysis of gene families and essential genes.

A majority of the knock-out mutants reported so far were derived from strain B05.10 (see description in Tudzynski and Siewers, 2004). This strain is highly virulent on several host plants and is genetically stable. As it consistently yields high transformation rates, it is now one of the standard recipient strains in most laboratories and it was also used for the first genome sequencing project (see above). Several recent investigations showed that the role of specific genes can differ between *B. cinerea* strains, e.g. targeted mutation of the pectin methylesterase *Bcpme1* reduced virulence in strain Bd90 (Valette-Collet *et al.*, 2003) but not in strain B05.10 (Kars *et al.*, 2005b). The ability to synthesize the toxin botrydial contributes to virulence in strain T4 but not in B05.10, probably due to the ability of the latter strain

to produce a second class of toxins, botcinolides (Siewers *et al.*, 2005). Mutation of the *BOs1* gene encoding a histidine kinase in strain B05.10 leads to an inability to penetrate (W. Liu and S. Fillinger, unpublished data), whereas the same mutant in strain UWS 111 still forms normal appressoria and is able to penetrate (Viaud *et al.*, 2006). These data emphasize the importance of the choice of strain to be used in such experiments and the urgent need to standardize these parameters in the *Botrytis* research community. On the other hand, these data show the high degree of flexibility of the pathogen. From this point of view it is interesting to have genome sequences at hand of two strains (T4 and B05.10) that differ considerably in virulence.

Unbiased gene identification

Since the classical 'candidate genes' have more or less been functionally analysed in *Botrytis*, unbiased cloning approaches gain interest because they offer the perspective of identifying genes with novel functions. Two general strategies are applied. First, screening can be performed for genes that are differentially expressed, e.g. specifically *in planta* or in certain developmental stages, and these genes can subsequently be analysed for their contribution to virulence by targeted inactivation. Techniques for differential gene expression studies which have been successfully applied in *B. cinerea* comprise differential screening of cDNA libraries, differential display RT-PCR (Benito *et al.*, 1996), suppression subtractive hybridization (Schulze Gronover *et al.*, 2004) and macroarrays (Chagué *et al.*, 2006; Viaud *et al.*, 2003). In the near future, microarrays will come available and allow whole-genome screening approaches.

A second strategy for identifying virulence genes in an unbiased manner comprises the use of random insertional mutagenesis, which has the distinct advantage of beginning from a known phenotype. While the classical REMI technique that was successfully applied in several phytopathogens (e.g. Balhadère *et al.*, 1999) did not perform well in *B. cinerea* (see Tudzynski and Siewers, 2004), *Agrobacterium*-mediated transformation has been established for *B. cinerea* in two laboratories (Rolland *et al.*, 2003; N. Segmüller and P. Tudzynski, unpublished data). This system fulfils two major criteria that are essential for use in insertional mutagenesis: most transformants carry single-copy integrations and the integration sites appear to be random. The availability of *B. cinerea* genomic sequences considerably facilitates the identification of the tagged genes. An insertional mutant library has been established and of the 2800 mutants derived so far, more than 30 are significantly impaired in virulence (S. Giesbert *et al.*, unpublished data)—a larger collection of virulence mutants than obtained so far with the targeted inactivation approach!

The availability of molecular tools and the option to use 'omics' approaches on a large scale in the near future attracts

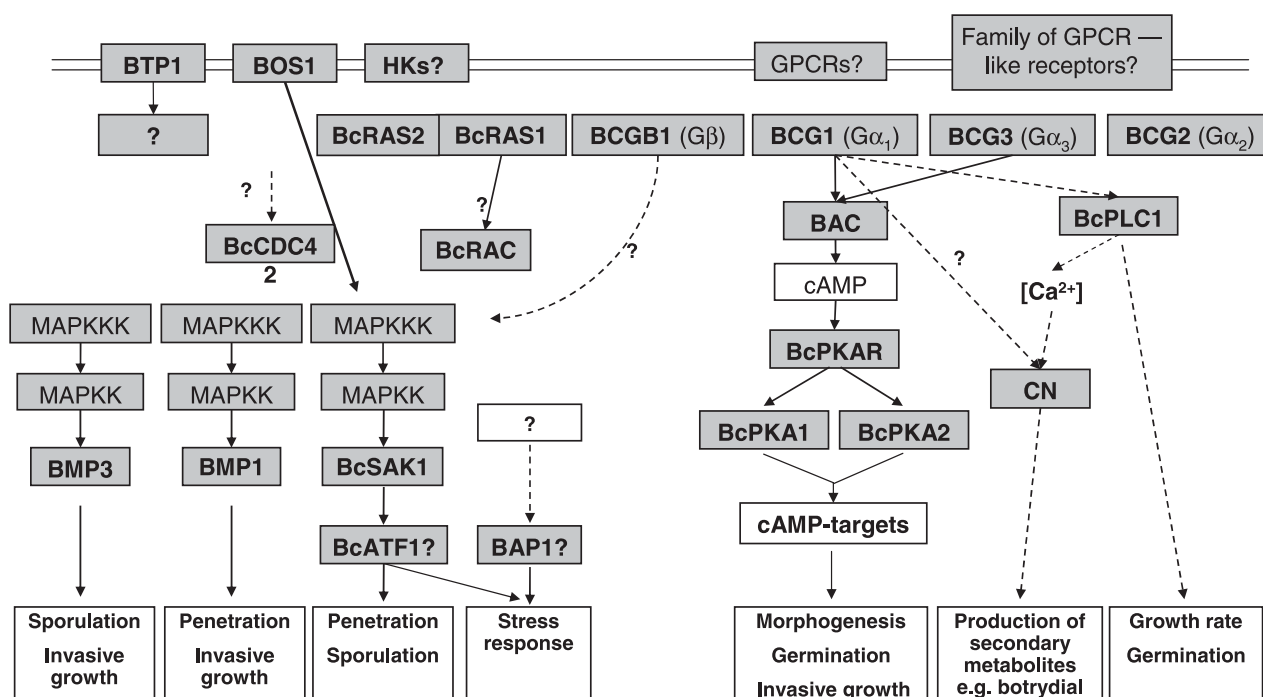


Fig. 3 Schematic representation of signalling pathways in *Botrytis cinerea*. Components in bold characters represent genes that are under functional investigation. For abbreviations of signalling components, see the main text.

more research groups, which in turn help to generate new techniques—a self-stimulating process. Thereby, *B. cinerea* is becoming the most extensively studied necrotrophic pathogen. Below we discuss the current status of functional analyses of genes involved in pathogenesis.

PERCEPTION OF ENVIRONMENT AND SIGNALLING DURING PATHOGENESIS

Sensing the environment and ensuring appropriate cellular responses are crucial challenges confronted by fungal pathogens and all living organisms in general. Failure at any step of signal sensing, transduction or cellular responses leads to abnormal growth and differentiation. Conserved signal transduction pathways, such as the cAMP-dependent and several MAP kinase pathways, have been shown to be important for most cell functions during morphogenesis, differentiation and pathogenic interactions. Recently, significant progress was made in characterization of several signalling components in *B. cinerea*, allowing the unravelling of the complicated regulatory networks in this pathogen. Since the last review of this field (Tudzynski and Schulze Gronover, 2004) several new genes involved in signalling processes affecting pathogenicity have been studied, and these

will be summarized here. The different pathways that are being studied and the processes in which these pathways operate are schematically represented in Fig. 3.

Components of the cAMP-dependent pathway

The cAMP-dependent signalling pathway is involved in multiple processes in plant-pathogenic fungi, including growth, conidiation and spore germination, nutrient sensing and virulence (Kronstad, 1997). In *B. cinerea*, the components of this pathway are fully characterized or under investigation. Among them are three genes encoding Gα subunits of heterotrimeric G-proteins named BCG1, BCG2 (Schulze Gronover *et al.*, 2001) and BCG3 (Döhlemann *et al.*, 2006), the adenylate cyclase-encoding gene *bac* (Klimpel *et al.*, 2002), genes for two catalytic subunits (*bcpka1* and *bcpka2*) and the regulatory subunit (*bcpkar*) of the cAMP-dependent protein kinase (PKA) (C. Huesmann and B. Tudzynski, unpublished data).

Deletion of each of these genes individually resulted in impaired virulence, but never to a total loss of pathogenicity. A very pronounced effect was observed in the $\Delta bcbg1$ mutant, which was able to produce conidia and penetrate plant tissue, but lesion development was fully arrested at this stage. Expansion of

lesions and soft rot development by $\Delta bcbg1$ mutants have never been observed (Schulze Gronover *et al.*, 2001). Both $\Delta bcbg2$ and $\Delta bcbg3$ mutants were able to invade the plant but spreading lesion formation was delayed compared with the wild-type. The $\Delta bcbg3$ mutant showed reduced conidial production and an impaired sugar-induced germination, which may account for the delayed infection process (Döhlemann *et al.*, 2006). While the phenotype of the $\Delta bcbg3$ mutant is almost completely restored by supplementation with cAMP, several functions of BCG1 seem to be cAMP-independent. Thus, addition of cAMP to the $\Delta bcbg1$ mutant restored the wild-type colony morphology but not the loss of protease secretion and production of the phytotoxin botrydial, suggesting that BCG1 controls at least one additional signalling pathway. The adenylate cyclase BAC is activated by two G α subunits, BCG1 and BCG3, as concluded from the observation that the Δbac mutant showed phenotypic similarities to both the $\Delta bcbg1$ and the $\Delta bcbg3$ mutants. Both the $\Delta bcbg1$ and the Δbac mutants form compact colonies on high sucrose-containing medium (Klimpel *et al.*, 2002), whereas the $\Delta bcbg3$ and Δbac mutants both showed an impaired spore germination (Döhlemann *et al.*, 2006). In contrast to the $\Delta bcbg3$ mutant, the Δbac mutant was unable to sporulate *in planta*, while *in vitro* conidiation was unaffected (Klimpel *et al.*, 2002).

Recently, the genes encoding the two catalytic and the regulatory subunits of the PKA were cloned and deletion mutants are under investigation (C. Huesmann and B. Tudzynski, unpublished data). The $\Delta bcpka1$ mutants displayed the most pronounced phenotypes *in vitro*; they grew slowly and produced only small colonies on different complete and synthetic media. As for the Δbac mutants, the development of spreading lesions by $\Delta bcpka1$ mutants was delayed and soft rot of whole leaves never occurred. In contrast to the Δbac mutant, the $\Delta bcpka1$ mutants are able to sporulate *in planta*. A strain mutated in the $bcpka2$ gene (encoding the second catalytic subunit of the PKA) showed wild-type growth, conidiation, germination and infection (C. Huesmann and B. Tudzynski, unpublished data).

Besides these components of the cAMP-dependent pathway, the gene for the G β -subunit ($bcbg1$) of the heterotrimeric G-protein has been cloned and deleted. The $\Delta bcbg1$ mutants showed altered colony morphology, delayed and reduced conidiation, and delayed penetration of plant tissue. Infection was arrested at the stage of secondary lesion formation, preventing soft rot development (J. Schumacher and B. Tudzynski, unpublished data). So far, little overlap has been found in the *B. cinerea* genes that are regulated by BCG1 (G α) and BCG1 (G β). In contrast, in the chestnut pathogen *Cryphonectria parasitica* the transcripts of c. 100 genes showed altered (either induced or repressed) expression levels in both the G α mutant $\Delta cpg1$ and the G β mutant $\Delta cpgb1$. In most cases, these transcripts appeared to be co-regulated, suggesting a considerable redundancy in pathway control or extensive cross-talk between the G α and G β subunit-

controlled pathways (Dawe *et al.*, 2004). These differences between *B. cinerea* and *C. parasitica* illustrate that signalling components that are highly conserved in fungal pathogens may act in very different ways.

MAP kinase-controlled signalling pathways

It has been shown for several plant pathogens that MAP kinases, especially the homologues of the *Magnaporthe grisea* PMK1 (Xu and Hamer, 1996; Xu, 2000), are essential for the early phases of infection, specifically the penetration of plant surfaces (Jenczmi-onka and Schäfer, 2005; Mey *et al.*, 2002; Solomon *et al.*, 2005). In *B. cinerea*, deletion of the *pmk1*-homologous gene, *bmp1*, resulted in altered growth rate, reduced conidiation and total inability to penetrate host tissue (Zheng *et al.*, 2000). Recently, the same gene has been deleted in a second *B. cinerea* wild-type strain, B05.10 (Döhlemann *et al.*, 2006). While the new $\Delta bmp1$ mutants remained unable to penetrate plant tissue and still produce fewer conidia, they showed less pleiotropic growth defects *in vitro*. However, detailed analysis of the $\Delta bmp1$ mutants, in the genetic background of strain B05.10, revealed a role of BMP1 in carbon source-induced germination of conidia in addition to the previously described phenotypes (Döhlemann *et al.*, 2006).

Recently, a second MAP kinase-encoding gene, the homologue of the *Saccharomyces cerevisiae* HOG1, has been cloned and characterized. While the yeast HOG1 kinase is mainly activated by hyperosmotic stress, the homologues in filamentous fungi may also be involved in responses to oxidative stress or fungicides. Deletion of *hog1*-like genes in pathogenic fungi, such as *M. grisea*, *Colletotrichum lagenarium* and *C. parasitica*, did not or only slightly affected pathogenicity (Dixon *et al.*, 1999; Kojima *et al.*, 2004; Park *et al.*, 2004). The *B. cinerea* HOG1 homologue, named BcSAK1, shows unique functional features: it is phosphorylated when *B. cinerea* is exposed to certain fungicides, osmotic stress and oxidative stress. The $\Delta bcsak1$ mutants are significantly impaired in vegetative and pathogenic development. They fail to produce conidia, show increased sclerotial development and are unable to penetrate unwounded plant tissues (Segmüller *et al.*, 2007). This is by far the strongest phenotype associated with a stress-activated MAP cascade in phytopathogenic fungi. To study the impact of stress on pathogenic development in detail, homologues of yeast transcription factors involved in stress response are currently being characterized. A homologue of the *S. cerevisiae* *yap1* gene, *bap1*, was functionally analysed. The $\Delta bap1$ mutants were more sensitive to oxidative stress *in vitro*, but showed normal virulence. Northern analyses showed that BAP1 controls several typical oxidative stress response genes, but these genes differ from the ones regulated by BcSAK1, indicating that BAP1 perhaps acts independently from the BcSAK1 cascade (N. Temme and P. Tudzynski, unpublished data). The *B. cinerea*

genes encoding homologues of the yeast response regulator Skn7 and the bZIP factor ATF1 (which acts in yeast downstream of the HOG cascade) are currently being functionally analysed (N. Temme and P. Tudzynski, unpublished data).

B. cinerea, as with most other fungi, contains a third MAP kinase-encoding gene, designated *bmp3* encoding the SlT2 homologue of *S. cerevisiae*. Deletion of the *bmp3* gene led to reduced vegetative growth on media with low osmolarity, impaired conidiation and failure to form sclerotia (Rui and Hahn, 2007). Although some defects in this mutant are similar to those in other pathogenic fungi (e.g. the impaired ability to invade plant tissue), BMP3 has features unique for a SlT2-type MAP kinase, such as the low osmolarity-induced growth inhibition.

The Ca²⁺/calmodulin-dependent signalling pathway

The role of calcineurin phosphatase and cyclophilin A, highly conserved components of the Ca²⁺/calmodulin-dependent signalling pathway, was investigated in *B. cinerea* (Viaud *et al.*, 2003). Immuno-suppressive drugs, such as cyclosporin A (CsA) and FK506, inhibit calcineurin activity by binding to the peptidyl-prolyl isomerases cyclophilin A and FKBP12, respectively. The protein–drug complexes bind to the hydrophobic interface between both subunits thus inhibiting the calcineurin phosphatase (reviewed by Kraus and Heitman, 2003).

In *B. cinerea*, the genes encoding the cellular targets of both drugs, the cyclophilin A- and FKBP12-encoding genes *bcp1* and *bcpic5*, respectively, have been deleted yielding drug-resistant mutants that are affected in virulence on bean and tomato leaves (Gioti *et al.*, 2006; Viaud *et al.*, 2003). Targeted disruption of the calcineurin gene was unsuccessful, however, probably because such mutants are lethal. Therefore, calcineurin was inhibited using CsA and cDNA from mycelia treated or non-treated with CsA was used for differential hybridization of macroarrays. This approach allowed the identification of 18 calcineurin-dependent (*CND*) genes among 2839 *B. cinerea* genes which were down-regulated by CsA. Among the co-regulated *CND* genes, three were shown to be organized as a physical cluster that could be involved in secondary metabolism (Viaud *et al.*, 2003), which later appeared to be required for botrydial biosynthesis (Siewers *et al.*, 2005).

As previously mentioned, $\Delta bcp1$ deletion mutants lost the ability to produce the phytotoxic secondary metabolites botrydial and botcinolides (Schulze Gronover *et al.*, 2001, 2004). By using a cDNA macroarray approach, the co-regulation of several genes by BCG1 and calcineurin has recently been shown, confirming the interconnection between these signalling pathways (J. Schumacher *et al.*, unpublished data). Additional components of the Ca²⁺/calmodulin-dependent signalling pathway, such as the transcription factor CRZ1 and phospholipase C, are currently under investigation (J. Schumacher and B. Tudzynski, unpublished data).

Small G-proteins

Ras- and Rho-type GTPases of the RAS superfamily have been examined in a wide range of eukaryotes and often play overlapping roles in cell polarization, differentiation and development. As with other filamentous fungi, *B. cinerea* contains two proteins of the Ras subfamily, BcRAS1 and BcRAS2, and the encoding genes have been cloned and deleted. $\Delta bcras1$ mutants were viable but displayed profound growth defects. They showed an irregular hyphal morphology and lost their ability for polarized growth. In addition, $\Delta bcras1$ mutants did not produce spores and were totally non-pathogenic. The phenotype of $\Delta bcras2$ mutants was not as strong, having slightly impaired spore formation and pathogenicity, and decreased growth rates on solid media (L. Kokkeling *et al.*, unpublished data).

Unlike yeast, filamentous fungi have Rac-like proteins from the Rho subfamily in addition to Cdc42 (Boyce *et al.*, 2005). In *B. cinerea*, $\Delta bcrac$ mutants displayed a similarly strong phenotype as $\Delta bcras1$ mutants (loss of polarized growth, spore formation and pathogenicity), suggesting that BcRAS1 and BcRAC act in the same signalling pathway. In contrast, growth and pathogenic development in $\Delta bccdc42$ mutants are only slightly affected (L. Kokkeling and P. Tudzynski, unpublished data).

Beside these genes, several members of the Rab subfamily of small GTPases have been cloned. In eukaryotic cells, Rab-like GTPases are major regulators of vesicular trafficking and are involved in essential processes including exocytosis, endocytosis and cellular differentiation (Siriputthaiwan *et al.*, 2005). So far, *bcrab2* in *B. cinerea* has been characterized in more detail. $\Delta bcrab2$ mutants can penetrate plant tissue and develop small lesions but spreading lesions have never been observed. These results indicate that the Rab/GTPase BcRAB2 is essential for invading host cells, probably by regulating the intracellular transport of secretory vesicles involved in the delivery of proteins to the extracellular medium (P. Hantsch and B. Tudzynski, unpublished data).

Sensors and receptors

Fungi respond to a variety of environmental signals that regulate metabolism and development as well as interactions with hosts. Cell-surface receptors perceive these signals and relay them to intracellular signalling pathways. The growing number of sequenced fungal genomes allows a better understanding of classes of proteins that could be involved in signal reception. In particular, families of G-protein-coupled receptors (GPCRs) have attracted major attention in plant pathogenic fungi (DeZwaan *et al.*, 1999; Kulkarni *et al.*, 2005). Two GPCR subfamilies can be distinguished in fungi on the basis of the presence or absence of an amino-terminal extracellular cysteine-rich EGF-like domain (CFEM domain) that is characteristic in human GPCRs. A prototype representative of a fungal GPCR with such a domain is the

M. grisea PTH11. In *M. grisea*, 61 PTH11-related proteins were identified, thereby constituting the largest number of GPCR-like proteins reported in fungi to date (Kulkarni *et al.*, 2005). In *B. cinerea*, so far only one gene, *btp1*, encoding a protein with seven transmembrane domains and significant homology to PTH11, has been found by an SSH approach (Schulze Gronover *et al.*, 2005). However, the protein does not contain the CFEM domain, suggesting that it belongs to the first class of GPCRs. The $\Delta btp1$ mutants were only slightly impaired in virulence and BTP1 thus probably does not interact with BCG1 during pathogenesis.

Beside GPCRs, two-component histidine kinases (HKs) are proteins by which organisms sense extracellular signals and adapt to their environment. In response to a specific signal, the HK auto-phosphorylates a conserved histidine residue. The phosphate is then transferred to a conserved aspartic acid residue in a response regulator (RR) protein, resulting in changed transcription or regulation of a MAP kinase cascade (Wolanin *et al.*, 2002). The *B. cinerea* genome sequence revealed 20 HKs in 11 classes (Catlett *et al.*, 2003). The class III HK, BOS1, was shown to be involved in osmoregulation, resistance to dicarboximide and phenylpyrrole fungicides and virulence. Interestingly, $\Delta bos1$ mutants displayed a phenotype similar to that of $\Delta bcsak1$ mutants (loss of conidiation, osmosensitivity, and resistance to fungicides). However, in contrast to $\Delta bcsak1$ mutants, $\Delta bos1$ strains can still penetrate, but are significantly reduced in the ability to invade host cells (Viaud *et al.*, 2006). The difference in penetration capacity is probably due to the use of different strains as recipients, since deletion of the same gene in B05.10 led to the same penetration defect as in $\Delta bcsak1$ (W. Liu and S. Fillinger, personal communication). These data indicate that BOS1 is the major upstream component of the BcSAK1 cascade.

Recently, a member of class X HKs, homologous to *Schizosaccharomyces pombe* Mak2/3 which is involved in oxidative stress responses (Buck *et al.*, 2001), has been functionally analysed in *B. cinerea*. The mutant is impaired in growth on 5 mM H₂O₂, suggesting that this phospho-relay system is involved in oxidative stress response caused by low doses, whereas the BcSAK1 cascade is required for responses to high doses of H₂O₂ (N. Temme and P. Tudzynski, unpublished data). Deletion of the *bchk5* gene, encoding a homologue of the single HK in *S. cerevisiae*, SLN1, showed no obvious phenotype. This result was unexpected as SLN1 is the upstream effector of the HOG1-cascade in yeast (Y. Cuesta Arenas and J. van Kan, unpublished data).

FUNGAL EFFECTOR MOLECULES INVOLVED IN PATHOGENESIS

The sensing and signalling processes described above ultimately lead to the production of effector molecules that enable *B. cinerea* to kill the host and decompose the plant tissue in order to convert it into fungal biomass.

Enzymatic determinants

Pathogens landing on a leaf must penetrate the host surface, composed of cutin covered with wax. *B. cinerea* differentiates appressoria that breach the cuticle by means of a penetration peg (Tenberge, 2004). *B. cinerea* appressoria require for penetration a membrane-associated protein BcPLS1 (Gourgues *et al.*, 2004), homologous to a protein that was shown to be essential for appressorium function in *M. grisea* (Clergeot *et al.*, 2001). *B. cinerea* *Bcpls1*-deficient mutants form appressoria of normal structural appearance, but they cannot penetrate an intact plant surface (Gourgues *et al.*, 2004), for reasons that remain unclear.

B. cinerea appressoria presumably secrete enzymes to breach the plant surface. The role of a cutinase and a lipase were studied. Deletion of a cutinase gene and a lipase gene, either separately or together, did not detectably reduce virulence (van Kan *et al.*, 1997; Reis *et al.*, 2005). The genome of *B. cinerea*, however, contains multiple additional cutinase and lipase genes and unravelling the role of these enzyme families in pathogenesis requires further study. One fungal enzyme that plays an important role in host surface penetration by appressoria is a secreted superoxide dismutase (BcSOD1) that is active during cuticle penetration by the appressorium (Rolke *et al.*, 2004). An oxidative burst occurs during cuticle penetration (Schouten *et al.*, 2002) and BcSOD1 may contribute to this process. Deletion of the *Bcsod1* gene led to reduced virulence on multiple hosts (Rolke *et al.*, 2004). The source of superoxide acting as substrate for BcSOD1 remains to be identified. Recently, homologues of the enzymes involved in oxidative burst generation in animals and plants, NADPH oxidases, have also been identified in fungi (for a review see Aguirre *et al.*, 2005). These enzymes would be good candidates for reactive oxygen species (ROS)-generating systems. Indeed, functional analysis of the two *nox* genes present in *B. cinerea* (*bcnox1*, *bcnox2*) showed that they both have significant impact on virulence (N. Segmüller and P. Tudzynski, unpublished data). However, their impact on the ROS status *in planta* has yet to be determined.

Upon breaching the cuticle, the penetration peg often grows into the anticlinal wall of the underlying epidermal cell, which is rich in pectin. The invasion of this layer therefore involves the action of pectinases, especially the endopolygalacturonase BcPG2. Mutants in which the *Bcpg2* gene was deleted showed a delay in primary lesion formation on bean and tomato leaves (Kars *et al.*, 2005a), while mutants in other endopolygalacturonase genes did not show such a delay. *B. cinerea* contains at least six endopolygalacturonase genes (Wubben *et al.*, 1999), of which the expression during host infection varies depending on the plant species, tissue type and incubation conditions applied (ten Have *et al.*, 2001), suggesting a degree of functional versatility. Deletion of two endopolygalacturonase genes, separately, resulted in a pronounced reduction of virulence on several host plants (ten

Have *et al.*, 1998; Kars *et al.*, 2005a), whereas deletion of the other four endopolygalacturonase genes had no notable effect on virulence (I. Kars and J. A. L. van Kan, unpublished data).

The role of pectin methylsterases (PMEs) in pathogenesis was also studied. It is considered that endopolygalacturonases cannot efficiently depolymerize highly methylated pectin, hence demethylation by PMEs presumably precedes and facilitates the action of endopolygalacturonases. This predicts that PMEs are important for fungal growth on highly methylated pectin as sole carbon source and for virulence on plant tissues with highly methylated pectin (such as leaves), but not on tissues with low pectin methylation (such as fruit). The phenotype of single and double mutants in two *Bcpme* genes in strain B05.10, however, was not different from the wild-type (Kars *et al.*, 2005b). Surprisingly, the wild-type strain and the *Bcpme*-deficient mutants even grew better on 75% methylated pectin than on non-methylated polygalacturonic acid, suggesting that pectin demethylation by PMEs is not important for its depolymerization *in vivo* by endopolygalacturonases (Kars *et al.*, 2005b).

Besides pectinases, other types of cell-wall-degrading enzymes produced by *B. cinerea* have been studied. Deletion of a cellulase gene did not affect virulence (Espino *et al.*, 2005), whereas the deletion of a β -1,4-xylanase gene delayed lesion formation and reduced lesion outgrowth by more than 70% (Brito *et al.*, 2006).

Phytotoxic compounds

B. cinerea can produce a spectrum of phytotoxic metabolites of low molecular weight, as well as phytotoxic proteins. The best studied phytotoxic metabolite is botrydial (Colmenares *et al.*, 2002). Botrydial was initially identified in liquid cultures but spectroscopic methods allowed detection of its accumulation in infected plant tissue, at concentrations above the toxicity threshold (Deighton *et al.*, 2001). The biosynthetic pathway for botrydial has been resolved biochemically (Colmenares *et al.*, 2002). A number of genes involved in its synthesis were identified, which are organized in a cluster containing at least two cytochrome P450 monooxygenase genes as well as a terpene cyclase gene (Siewers *et al.*, 2005). Deletion of one of the cytochrome P450 monooxygenase genes, named *Bcbot1*, in three different strains resulted in severe reduction of virulence in one strain, but not in the others (Siewers *et al.*, 2005), indicating that certain strains might strictly rely on botrydial to kill host cells, while others can produce additional toxins, such as botcinolides (Reino *et al.*, 2004). The biosynthetic pathway of botcinolide remains to be resolved.

Besides phytotoxic metabolites, *B. cinerea* can produce at least three distinct phytotoxic proteins, i.e. two NEP1-like proteins (Staats *et al.*, 2007) and a Snodprot homologue named *Bcsp1* (Chagué *et al.*, 2006; Kunz *et al.*, 2006). *B. cinerea* NEP1-like proteins are able to cause host cell death both by a combination

or mixture of apoptotic and necrotic mechanisms (A. Schouten *et al.*, unpublished data). The role of phytotoxic proteins in pathogenesis of *B. cinerea* is still under investigation.

In addition, *B. cinerea* is a notorious producer of oxalic acid *in vitro* (Germeier *et al.*, 1994) and *in planta* (Verhoeff *et al.*, 1988). Oxalic acid may be a cofactor in pathogenesis rather than a primary phytotoxic agent. Culturing *B. cinerea* in low ambient pH resulted in the enhanced production of various secreted enzymes that have an optimal activity at low pH and are thus stimulated by simultaneous secretion of oxalate (Manteau *et al.*, 2003). Moreover, oxalate may stimulate pectin hydrolysis resulting from endopolygalacturonase action by sequestering Ca^{2+} ions from (intact or partially hydrolysed) Ca-pectates in the cell walls. The removal of Ca^{2+} ions disturbs intermolecular interactions between pectic polymers and disrupts the integrity of the pectic backbone. Consequently, the pectic matrix absorbs water and swells, as observed by Mansfield and Richardson (1981). *B. cinerea* produces oxalate *in vitro* by means of oxaloacetate hydrolase (*Bcoah1*), an enzyme converting oxaloacetate into pyruvate and oxalate. Mutants in the *Bcoah1* gene are defective in oxalate production *in vitro* (Han *et al.*, 2007) and they retained their ability to produce sclerotia (J. A. L. van Kan, unpublished), unlike oxalate non-producing mutants of *Sclerotinia sclerotiorum* (Godoy *et al.*, 1990). The effect of the deletion of the *Bcoah1* gene on pathogenesis is under investigation (J. van Kan and K. Plummer, unpublished data).

INFECTION REQUIRES ACTIVE PARTICIPATION BY THE HOST

Until a few years ago, it was considered that host plants played a rather passive role in the interaction with necrotrophic pathogens. Recent research has revealed that the host plays a much more active role than previously anticipated and the interactions between plants and necrotrophic fungi are in fact more subtle than previously appreciated. The ability to induce programmed cell death appears to play a pivotal role in the success of *B. cinerea*.

Cuticle penetration and primary lesion formation by *B. cinerea* triggers an oxidative burst, both in the plant plasmamembrane and in the extracellular sheath covering the surface of fungal hyphae (Schouten *et al.*, 2002; Tenberge, 2004). Histochemical staining in *Botrytis*-infected Arabidopsis leaves revealed that the oxidative burst in a compatible interaction comprises the simultaneous production of hydrogen peroxide and nitric oxide, as well as the formation of proteolytic, autophagosome-like vesicles at the host–pathogen interface (van Baarlen *et al.*, 2007). *B. cinerea* infection leads to the accumulation of free radicals, both at the host–pathogen interface and at some distance from the invading hyphae (Muckenschnabel *et al.*, 2001a, 2003) culminating in lipid peroxidation (Deighton *et al.*, 1999; Muckenschnabel

et al., 2001b, 2002) and depletion of antioxidants (Muckenschnabel *et al.*, 2002). Altogether, these oxidative processes cause massive perturbation of the redox status in and around the infected tissue, thereby promoting disease progress (Lyon *et al.*, 2004). Besides the fungal secreted superoxide dismutase BcSOD1, the plant enzyme that most prominently contributes to the oxidative burst is the plasma membrane-associated NADPH-dependent oxidase.

The infection of Arabidopsis by *B. cinerea* induces cell death concomitant with nuclear condensation and expression of the HR-specific gene *Hsr203* (Govrin and Levine, 2000). In *B. cinerea*-infected tomato, expression of *Hsr203* and activation of metacaspase activity were observed, both indicative of the occurrence of programmed cell death (Hoerberichts *et al.*, 2003). Growth of *B. cinerea* in Arabidopsis was suppressed in the HR-deficient mutant *dnd1* and was stimulated by HR triggered by simultaneous inoculation with an avirulent bacterium (Govrin and Levine, 2000). Van Baarlen *et al.* (2007) performed inoculation experiments on a collection of 12 Arabidopsis knockout mutants affected in metacaspase or vacuolar processing enzyme genes, involved in cell death and senescence processes. Generally, mutations that promoted cell death increased susceptibility, while mutations that delayed cell death increased resistance to *B. cinerea*. Changes in susceptibility in these mutants were significant but small, presumably due to the large functional redundancy within the metacaspase and VPE gene families. A model was proposed in which the balance between life and death is an important determinant for the outcome of the Arabidopsis–*B. cinerea* interaction (van Baarlen *et al.*, 2007). The observation that programmed cell death is an important determinant in the interaction of *B. cinerea* with its host plants is supported by the fact that transgenic plants expressing heterologous anti-apoptotic genes have an increased resistance to *Sclerotinia sclerotiorum* and *B. cinerea* (Dickman *et al.*, 2001). It remains to be established whether the phytotoxic metabolites and proteins produced by *B. cinerea* (discussed above) are inducers of programmed cell death, rather than direct-acting toxins causing disorganized death (necrosis).

HOST DEFENCE SYSTEMS

Throughout the course of an interaction between *B. cinerea* and its host, the plant vigorously attempts to prevent pathogen invasion and outgrowth by activating multiple defence pathways, including the production of antifungal metabolites and pathogenesis-related proteins (reviewed by van Baarlen *et al.*, 2004). *B. cinerea* infection of tomato and Arabidopsis induces the expression of multiple genes encoding defence-related proteins that are considered to be markers for defence pathways governed by salicylic acid, ethylene and jasmonate (Benito *et al.*, 1998; Díaz *et al.*, 2002; Thomma *et al.*, 2001). It is likely that homologous genes and similar defence pathways are induced in other host

plants, but this is less extensively documented. Phytohormone-mediated defence pathways contribute to basal resistance to *B. cinerea*, as mutants in these pathways showed a partial, sometimes dramatic, increase in susceptibility to grey mould (Audenaert *et al.*, 2002; Díaz *et al.*, 2002; Ferrari *et al.*, 2003; Thomma *et al.*, 1998, 1999). A recent genome-wide analysis of changes in the transcriptome of *B. cinerea*-infected Arabidopsis revealed a set of 621 up-regulated transcripts, of which only a third are under control of ethylene, jasmonate or salicylic acid-mediated signalling pathways (AbuQamar *et al.*, 2006). This study identified 30 genes encoding transcription factors which were up-regulated by infection and might be involved in regulating basal resistance against *B. cinerea*. Indeed, Arabidopsis mutants in which the genes encoding transcription factors R2R3MYB, WRKY70 and ZFAR1 were inactivated showed enhanced susceptibility to *B. cinerea* (Mengiste *et al.*, 2003; AbuQamar *et al.*, 2006), although mechanisms underlying the phenotype remain to be resolved.

Plants can produce polygalacturonase inhibiting proteins (PGIPs), leucine rich repeat-containing proteins that may inhibit endopolygalacturonases of plant pathogenic and non-pathogenic fungi (reviewed by Juge, 2006) by direct physical interaction between the two proteins. PGIPs display *in vitro* specificity towards different fungal endopolygalacturonases (Leckie *et al.*, 1999). Expression of PGIPs from different sources in transgenic plants resulted in a quantitative increase of resistance to *B. cinerea* (Agüero *et al.*, 2005; De Lorenzo and Ferrari, 2002; Ferrari *et al.*, 2003; Joubert *et al.*, 2006; Powell *et al.*, 2000). Recent research has shown that *in vitro* studies of PGIP–PG interactions only partially reveal the potential of PGIPs for increasing resistance. It was generally considered that among the family of PGIPs described thus far, the most potent *in vitro* inhibitors would be the most beneficial proteins to express in plants for achieving optimal resistance to *B. cinerea*. However, a grapevine PGIP, VvPGIP1, was described that did not display any detectable *in vitro* interaction with the *B. cinerea* endopolygalacturonase BcPG2, yet the proteins interacted *in planta* and the VvPGIP1 conferred partial protection from damage inflicted by BcPG2 action (Joubert *et al.*, 2007).

To be a successful pathogen on multiple host species, *B. cinerea* must be able to cope with plant defence compounds. The Arabidopsis phytoalexin camalexin is one example of a potent antifungal compound that contributes to basal resistance to *B. cinerea*, as evidenced by the increased susceptibility of camalexin-deficient mutants (van Baarlen *et al.*, 2007; Kliebenstein *et al.*, 2005). Differences in aggressiveness between *B. cinerea* isolates were attributed partly to the ability to detoxify camalexin (Kliebenstein *et al.*, 2005). Other examples of the ability of *B. cinerea* to counteract the activity of antifungal plant metabolites are provided by the enzymatic degradation of alpha tomatin (Quidde *et al.*, 1999) and the active secretion of plant defence compounds by

ABC (ATP-binding cassette) or MFS (major facilitator super-family) transporters (reviewed by de Waard *et al.*, 2006).

DISEASE MANAGEMENT

Chemical control

In 35 years since the first commercial use of methyl benzimidazole carbamate (MBC)-generating fungicides, acceptance has grown that for each new chemical the risk of resistance arising in *B. cinerea* is strong if the product is applied repeatedly. Consequently, mixed spray programmes have been devised, ideally with each spray chosen from a different fungicide group, to reduce the risk of substantial field resistance arising and to keep below the permitted maximum residue level for each active ingredient. The problem arises, however, when some horticultural crops need protection over extended periods because of sequential flowering and fruiting. The molecular target sites of modern fungicides and the mechanisms of resistance are gradually becoming clear and such studies will be greatly facilitated when the complete *B. cinerea* genome is analysed and annotated.

The chemicals used for control of *B. cinerea* have recently been reviewed (Leroux, 2004). Five categories of fungicides are recognized, namely those affecting respiration, microtubule assembly, osmoregulation, sterol biosynthesis inhibitors and those whose toxicity is reversed by amino acids (Rosslénbroich and Stuebler, 2000). Several multisite toxicants affecting fungal respiration have been used against *B. cinerea* over a long period without substantial resistance developing in field populations (e.g. thiram, mancozeb, captan, dichlofluanid, tolylfluanid). The genes *Dic1* and *Dic2* (Pollastro *et al.*, 1996) confer limited resistance to dichlofluanid; cross-resistance to various dithiocarbamates has been identified among captan-resistant isolates (Leroux, 2004). MBC-generating fungicides that inhibit β -tubulin formation developed resistance rapidly (conferred by the *Mbc1* gene) and now have limited use against *B. cinerea* because they have long persistence and residues accumulate. Dicarboximides have been used extensively as botryticides although their primary target site is not known. They show activity against both conidia and mycelium by affecting sensitivity to osmotic stress. Resistance to this group of chemicals was identified as a single polymorphic gene *Daf1* (Farettra and Pollastro, 1991). Recently, a gene named *BcOS1* (Leroux *et al.*, 2002) or *BOs1* (Cui *et al.*, 2002) was cloned and found to be homologous with the *Neurospora crassa os1* gene. According to Cui *et al.* (2002) *Daf1* and *BOs1* correspond to the same gene. The linker region of the *os1*-type histidine kinase could be the target site for dicarboximides (Leroux, 2004) and this is supported by the work of Cui *et al.* (2004). Dicarboximide-resistant isolates display a reduction in fitness as their frequency declines once spraying stops (Beever *et al.*, 1989; Raposo *et al.*, 2000). Strobilurin fungicides that inhibit cytochrome b control

B. cinerea and have the advantage that they are broad-spectrum fungicides potentially controlling several diseases. The anilinopyrimidines are useful botryticides that are antagonized by methionine and some other amino acids. These fungicides can prevent secretion of hydrolytic enzymes that play a role in pathogenesis, such as cutinases, lipases, cellulases and proteases (Miura *et al.*, 1994). Fenhexamid, a sterol biosynthesis-inhibiting fungicide, is the most recent and effective fungicide against *B. cinerea* (Rosslénbroich and Stuebler, 2000). Certain isolates from a defined *B. cinerea* subpopulation differ in their resistance to this fungicide *in vitro* (Albertini *et al.*, 2002; Fournier *et al.*, 2003).

Increased insensitivity of *B. cinerea* isolates to a combination of fungicides, also referred to as multi-drug resistance, is often associated with the action of ABC or MFS transporter families that transport molecules across the plasmamembrane (reviewed by de Waard *et al.*, 2006). Increased transcript levels of the *BcatrB* gene are observed in the presence of phenylpyrrole fungicides, but not dicarboximides, anilinopyrimidines and lanosterol 14 α -demethylase inhibitors (Schoonbeek *et al.*, 2001), whereas *BcatrD* is up-regulated in the presence of the latter three chemical groups (Hayashi *et al.*, 2001, 2002).

Biological control

Early studies on microbial ecology of the phyllosphere showed that there was considerable potential for use of microbial antagonists for control of *Botrytis* on crops. At least seven products have now been approved (Elad and Stewart, 2004) for use on food and non-food plants in greenhouses, under plastic tunnels or in the field in different countries. They have achieved niche markets in situations where heavy use of conventional fungicides has been restricted because of residues accumulating, or because of the restrictions imposed by importing countries. The original aspirations to deliver a single biological control agent (BCA) infrequently and then rely on its ability to self-disperse in a crop canopy to the required target zones has in most cases turned out to be unrealistically optimistic, but great advances have been made in the understanding of their biological modes of action. BCA formulations may include filamentous fungi such as *Trichoderma harzianum*, *Clonostachys rosea* (*Gliocladium roseum*) and *Ulocladium oudemansii*, the yeast *Candida oleophila*, or bacteria including *Streptomyces griseoviridis*, *Bacillus subtilis* and *Pseudomonas syringae*. Most BCAs are sprayed on the crop plant, but there has been some success in strawberries using bees with inoculation trays in hives to deliver *Gliocladium roseum* (Peng *et al.*, 1992), and *T. harzianum* (Bilu *et al.*, 2003, 2004; Kovach *et al.*, 2000). Compared with fungicides, BCAs often have restricted ranges of temperature or humidity for maximum microbial action, and they may be influenced by fluctuations in natural populations of phylloplane microbes responding to changes in plant exudates and the environment. Mixed microbial BCAs have been evaluated,

especially for controlling multiple post-harvest pathogens in apple and pear (Nunes *et al.*, 2002). For a full discussion of the modes of action and usefulness of BCAs against *Botrytis cinerea* see Elad and Stewart (2004).

Cultural practices

Grey mould is exacerbated by high humidity, reduced light and moderate temperature. Hence it is helpful in crop management to create an open canopy to provide adequate air movement and good light interception so that water droplets from rain or irrigation dry as soon as possible. High RH promotes conidial generation and allows germination and penetration of the host. Cultural practices that alleviate the effects of grey mould are diverse and often specific to particular species and cropping systems. In perennial woody plants, such as grapevines, pruning to reduce excessive vegetative growth of the plant has been shown to be beneficial (Gubler *et al.*, 1987). Excessive use of nitrogen fertilizer encourages rapid vegetative growth and increases the risk of grey mould and other diseases.

Some of the problems in soft fruit production caused by rainfall during the blossom period have been overcome by plastic rain shelters and tunnels, and facilitated a massive expansion in crop area for strawberries and raspberries. For example, 90% disease reductions in strawberries grown under plastic have been reported, compared with field-grown plants (Xiao *et al.*, 2001). However, it is still important to encourage ventilation to reduce high RH inside these structures and minimize wetting of foliage. When the plastic covers are removed in late summer there is still infection of leaves and stems, leading to over-wintering mycelium and sclerotia. Spectral modification of daylight by near-UV filters incorporated into plastic covers has been useful to reduce conidiation and infection in a number of crops (Reuveni and Raviv, 1992; Reuveni *et al.*, 1989; West *et al.*, 2000). In unheated greenhouses, the night temperature of plants can be lower than the air temperature due to irradiative cooling; heating briefly before sunrise to raise plant temperature above the ambient air temperature reduces dew formation on leaves and can control grey mould (Dik and Wubben, 2004).

Post-harvest management of fresh products relies extensively on 'cold-chain-marketing' of fruits harvested slightly under-ripe and with minimal wounding. Several plant defence systems still operate in the host tissues at this stage; if the temperature during shipment is strictly controlled, grey mould damage can be substantially reduced. In practice, the inoculum burden accumulating during the entire growing period greatly affects the spread of grey mould after harvest.

In the context of integrated crop management (IPM) there is great merit in using the maximum effort to reduce pesticide residues by minimal chemical treatment, alternating chemical groups to reduce resistance build-up; application of biological

control agent(s) appropriate for the temperature regime and humidities; scrupulous removal of dead crop material to remove inoculum; use of mulches to bury leaf litter and assist microbial breakdown of inoculum and conserve moisture; adequate plant spacing, effective pruning and good control of weeds to create open well-ventilated canopy; and management of insect pests that wound the plant and act as vectors. Disease forecasting, especially when combined with accurate local weather data, has been successful in reducing serious crop damage by specifying timely treatment in grape (Broome *et al.*, 1995) or strawberry (Berrie *et al.*, 2002).

Resistance breeding

Breeding for resistance against *B. cinerea* has been difficult and unrewarding in most crops, but recently there has been substantial advance in conventional breeding for grey mould resistance in tomato. The approaches taken for this plant may serve as a useful model in other plant families. Wild *Solanum* species closely related to cultivated tomato *Solanum lycopersicum* were found to display partial resistance in leaves and/or stems (Guimarães *et al.*, 2004; ten Have *et al.*, 2007). *S. habrochaites* genotype LYC4 was used for introgressing resistance to grey mould into *S. lycopersicum*. Three quantitative trait loci (QTLs) for resistance were identified in a segregating F2 population (Finkers *et al.*, 2007a). Seven additional QTLs were detected in an introgression line population consisting of 30 individual lines, each containing different well-defined segments of *S. habrochaites* LYC4 chromosomes in the genetic background of *S. lycopersicum* (Finkers *et al.*, 2007b). One of the genotypes obtained in these studies contained several QTLs and displayed a reduction of grey mould disease parameters as high as 85% compared with the susceptible parent (Finkers *et al.*, 2007a). As these studies were performed under rather high disease pressure, such partial resistance levels may possibly confer absolute resistance in normal greenhouse cultivation where lower disease pressure prevails. The QTLs for resistance to grey mould offer excellent perspectives for improved disease control in tomato. The mechanisms underlying the increased resistance remain to be unravelled and the introgression line population offers an excellent tool to study resistance mechanisms governed by the individual QTLs. With increasing understanding of the underlying mechanisms of genetic resistance it will be possible to use gene transfer techniques to enhance the host response to infection, without loss of other important plant characteristics required by agribusiness and the consumer.

PERSPECTIVES

In the last few years *Botrytis cinerea* has been adopted as an important model system in molecular phytopathology. Driven by the immense economic importance of this worldwide pathogen

and its special infection strategy, industrial and academic research groups have joined forces to unravel the biology of this necrotrophic pathogen, and to identify its potential weaknesses for development of new control strategies. Two separate major lines of research are being followed: the generation of resistant plants and the design of new, more specific and efficient fungicides. The infection strategy of *B. cinerea* includes the triggering of programmed cell death, which may hamper the design of plants with increased resistance without concomitant reduction of resistance against biotrophic pathogens. To unravel this complexity much basic research is necessary to understand better the role of different plant resistance pathways, and especially the methods by which *B. cinerea* interferes with, and exploits the plant defence systems to its own benefit (see discussion in van Baarlen *et al.*, 2007). Genome-wide transcriptome analyses (e.g. AbuQamar *et al.*, 2006) will help to identify key players in the host defence responses against *B. cinerea*, but data obtained from model systems like *Arabidopsis*, a non-natural host, will need to be substantiated by data from major host crop plants such as tomato or bean. In the long run, an increase of polygenic resistance to *B. cinerea* must have absolute priority over monogenic resistance obtained by introducing single genes (e.g. PGIPs) because the latter will probably break down quickly due to the genetic flexibility of the pathogen.

Although in recent years our understanding of the pathogenicity mechanisms of grey mould has significantly increased, there is no fungicide on the market that was developed on the basis of targeted molecular research, i.e. intelligent screening approaches. On the contrary, there appears to be a tendency in the big agricultural companies to lose patience and go back to the classical and less expensive techniques like chemical library screening. Such a retrograde step would be a mistake; industry should recognize the great potential resource arising in the rapidly accumulating knowledge of this model pathogen. The joint efforts of the increasing number of academic groups should be directed to the use of *B. cinerea* as a test bed for the feasibility of this molecular approach. Of course the great variability and the broad host range of this pathogen present a special technical challenge. The new genomic tools will help us to understand what really constitutes a necrotroph, especially with the possibility for comparisons between two closely related 'sister' necrotrophs, *B. cinerea* and *Sclerotinia sclerotiorum*, the white mould pathogen, and to compare their genomes both with related hemi-biotrophs and biotrophs, as well as more distantly related necrotrophs such as *Alternaria* spp. There are still a lot of 'black boxes': the exact penetration mechanisms; the role of secondary metabolites and of small proteinaceous toxins; the sensing and signalling mechanisms which allow an optimal adaptation to life inside a living plant; the role of stress adaptation and management of ROS; and especially the cross-talk between the established signal cascades. Finally, there is the problem of quiescence in the early

stages of infection. This aspect has not yet attracted much attention in molecular biology, but it could be very important in several crop plants, e.g. after flower infection of grapevine. Is this phenomenon just an efficient control of the fungus by plant defence systems, or does *B. cinerea* also have the capacity to behave as an endophyte in some hosts? Several surprises may emerge on the way to a full understanding of all facets and tactics that this pathogen can deploy.

ACKNOWLEDGEMENTS

We are grateful to Julia Schumacher for designing Fig. 3. We also thank P. Smith, Scottish Crop Research Institute, Invergowrie, Dundee, UK, for copy editing the draft text and Bart Thomma, Wageningen University, Laboratory of Phytopathology, for critical reading of the manuscript.

REFERENCES

- AbuQamar, S., Chen, X., Dhawan, R., Bluhm, B., Salmeron, J., Lam, S., Dietrich, R.A. and Mengiste, T. (2006) Expression profiling and mutant analysis reveals complex regulatory networks involved in *Arabidopsis* response to *Botrytis* infection. *Plant J.* **48**, 28–44.
- Agüero, C.B., Uratsu, S.L., Greve, C., Powell, A.L.T., Labavitch, J.M., Meredith, C.P. and Dandekar, A.M. (2005) Evaluation of tolerance to Pierce's disease and *Botrytis* in transgenic plants of *Vitis vinifera* L. expressing the pear PGIP gene. *Mol. Plant Pathol.* **6**, 43–51.
- Aguirre, J., Rios-Momberg, M., Hewitt, D. and Hansberg, W. (2005) Reactive oxygen species and development in microbial eukaryotes. *Trends Microbiol.* **13**, 111–118.
- Albertini, C. and Leroux, P. (2004) A *Botrytis cinerea* putative 3-keto reductase gene that is homologous to mammalian 17 β -hydroxysteroid dehydrogenase type 7 gene. *Eur. J. Plant Pathol.* **110**, 723–733.
- Albertini, C., Thebaud, G., Fournier, E. and Leroux, P. (2002) Eburicol 14 α -demethylase gene (*CVP51*) polymorphism and speciation in *Botrytis cinerea*. *Mycol. Res.* **106**, 1171–1178.
- Audenaert, K., De Meyer, G.B. and Höfte, M.M. (2002) Absciscic acid determines basal susceptibility of tomato to *Botrytis cinerea* and suppresses salicylic acid-dependent signaling mechanisms. *Plant Physiol.* **128**, 491–501.
- van Baarlen, P., Legendre, L. and van Kan, J.A.L. (2004) Plant defence compounds against *Botrytis* infection. In: *Botrytis: Biology, Pathology and Control*. (Elad, Y., Williamson, B., Tudzynski, P. and Delen, N., eds), pp. 143–161. Dordrecht, The Netherlands: Kluwer Academic Press.
- van Baarlen, P., Woltering, E.J., Staats, M. and van Kan, J.A.L. (2007) Histochemical and genetic analysis of host and non-host interactions of *Arabidopsis* with three *Botrytis* species: an important role for cell death control. *Mol. Plant Pathol.* **8**, 41–54.
- Backhouse, D. and Willets, H.J. (1984) A histochemical study of sclerotia of *Botrytis cinerea* and *Botrytis fabae*. *Can. J. Microbiol.* **30**, 171–178.
- Backhouse, D. and Willets, H.J. (1987) Development and structure of infection cushions of *Botrytis cinerea*. *Trans. Br. Mycol. Soc.* **89**, 89–95.
- Baldwin, T.K., Winnenburg, R., Urban, M., Rawlings, C., Koehler, J. and Hammond-Kosack, K.E. (2006) The pathogen–host interactions database (PHI-base) provides insights into generic and novel themes of pathogenicity. *Mol. Plant–Microbe Interact.* **19**, 1451–1462.

- Balhadère, P.V., Foster, A.J. and Talbot, N.J. (1999) Identification of pathogenicity mutants of the rice blast fungus *Magnaporthe grisea* by insertional mutagenesis. *Mol. Plant-Microbe Interact.* **12**, 129–142.
- Barnes, S.E. and Shaw, M.W. (2003) Infection of commercial hybrid primula seed by *Botrytis cinerea* and latent disease spread through the plants. *Phytopathology*, **93**, 573–578.
- Beever, R.E., Laracy, E.P. and Pak, H.A. (1989) Strains of *Botrytis cinerea* resistant to dicarboximide and benzimidazole fungicides in New Zealand vineyards. *Plant Pathol.* **38**, 427–437.
- Beever, R.E. and Weeds, P.L. (2004) Taxonomy and genetic variation of *Botrytis* and *Botryotinia*. In: *Botrytis: Biology, Pathology and Control*. (Elad, Y., Williamson, B., Tudzynski, P. and Delen, N., eds), pp. 29–52. Dordrecht, The Netherlands: Kluwer Academic Press.
- Benito, E.P., Prins, T.W. and van Kan, J.A.L. (1996) Application of differential display RT-PCR to the analysis of gene expression in a plant-fungus interaction. *Plant Mol. Biol.* **32**, 947–957.
- Benito, E.P., ten Have, A., van't Klooster, J.W. and van Kan, J.A.L. (1998) Fungal and plant gene expression during synchronized infection of tomato leaves by *Botrytis cinerea*. *Eur. J. Plant Pathol.* **104**, 207–220.
- Berrie, A.M., Harris, D.C. and Xu, X.M. (2002) A potential system for managing *Botrytis* and powdery mildew in main season strawberries. *Acta Hort.* **567**, 647–649.
- Bilu, A., Dag, A., Elad, Y. and Shaif, S. (2004) Honeybee dispersal of biocontrol agents: an evaluation of dispensing devices. *Biocontrol. Sci. Technol.* **14**, 607–617.
- Bilu, A., Dag, A., Shaif, S. and Elad, Y. (2003) Use of honeybees to disseminate *Trichoderma* (*Trichoderma harzianum* T39) to strawberry for the control of gray mold (*Botrytis cinerea*). *Phytoparasitica*, **31**, 296–297.
- Boyce, K.J., Hynes, M.J. and Andrianopoulos, A. (2005) The Ras and Rho GTPases genetically interact to co-ordinately regulate cell polarity during development in *Penicillium marneffei*. *Mol. Microbiol.* **55**, 1487–1501.
- Brito, N., Espino, J.J. and Gonzalez, C. (2006) The endo- β -1,4-xylanase Xyn11A is required for virulence in *Botrytis cinerea*. *Mol. Plant-Microbe Interact.* **19**, 25–32.
- Broome, J.C., English, J.T., Marois, J.J., Latorre, B.A. and Aviles, J.C. (1995) Development of an infection model for *Botrytis* bunch rot of grapes based on wetness duration and temperature. *Phytopathology*, **85**, 97–102.
- Buck, V., Quinn, J., Soto Pino, T., Martin, H., Saldanha, J., Makino, M., Morgan, B.A. and Millar, J.B.A. (2001) Peroxide sensors for the fission yeast stress-activated mitogen-activated protein kinase pathway. *Mol. Biol. Cell*, **12**, 407–419.
- Burgess, D.R., Bretag, T.W. and Keane, P.J. (1997) Seed to seedling transmission of *Botrytis cinerea* in chickpea and disinfestation of seed with moist heat. *Aust. J. Exp. Agric.* **37**, 223–229.
- Catlett, N.L., Yoder, O.C. and Turgeon, B.G. (2003) Whole-genome analysis of two-component signal transduction genes in fungal pathogens. *Eukaryot. Cell*, **2**, 1151–1161.
- Chagué, V., Danit, L.-V., Siewers, V., Schulze Gronover, C., Tudzynski, P., Tudzynski, B. and Sharon, A. (2006) Ethylene sensing and gene activation in *Botrytis cinerea*: a missing link in ethylene regulation of fungus-plant interactions? *Mol. Plant-Microbe Interact.* **19**, 33–42.
- Clergeot, P.H., Gourgues, M., Cots, J., Laurans, F., Latorse, M.-P., Pépin, R., Tharreau, D., Nottoghem, J.-L. and Lebrun, M.-H. (2001) PLS1, a gene encoding a tetraspanin-like protein, is required for penetration of rice leaf by the fungal pathogen *Magnaporthe grisea*. *Proc. Natl Acad. Sci. USA*, **98**, 6963–6968.
- Coertze, S., Holz, G. and Sadie, A. (2001) Germination and establishment of infection on grape berries by single airborne conidia of *Botrytis cinerea*. *Plant Dis.* **85**, 668–677.
- Cole, L., Dewey, F.M. and Hawes, C.R. (1996) Infection mechanisms of *Botrytis* species: pre-penetration and pre-infection processes of dry and wet conidia. *Mycol. Res.* **100**, 277–286.
- Coley-Smith, J.R., Verhoeff, K. and Jarvis, W.R. (1980) *The Biology of Botrytis*. London: Academic Press.
- Colmenares, A.J., Aleu, J., Durán-Patrón, R., Collado, I.G. and Hernández-Galán, R. (2002) The putative role of botrydial and related metabolites in the infection mechanism of *Botrytis cinerea*. *J. Chem. Ecol.* **28**, 997–1005.
- Cui, W., Beever, R.E., Parkes, S.L. and Templeton, M.D. (2004) Evolution of an osmosensing histidine kinase in field strains of *Botryotinia fuckeliana* (*Botrytis cinerea*) in response to dicarboximide fungicide usage. *Phytopathology*, **94**, 1129–1135.
- Cui, W., Beever, R.E., Parkes, S.L., Weeds, P.L. and Templeton, M.D. (2002) An osmosensing histidine kinase mediates dicarboximide fungicide resistance in *Botryotinia fuckeliana* (*Botrytis cinerea*). *Fung. Genet. Biol.* **36**, 187–198.
- Dawe, A.L., Segers, G.C., Allen, T.D., McMains, V.C. and Nuss, D.L. (2004) Microarray analysis of *Cryphonectria parasitica* Galpha- and Gbeta/gamma-signalling pathways reveals extensive modulation by hypovirus infection. *Microbiology*, **150**, 4033–4043.
- De Lorenzo, G. and Ferrari, S. (2002) Polygalacturonase-inhibiting proteins in defense against phytopathogenic fungi. *Curr. Opin. Plant Biol.* **5**, 295–299.
- Deighton, N., Muckenschnabel, I., Colmenares, A.J., Collado, I.G. and Williamson, B. (2001) Botrydial is produced in plant tissues infected by *Botrytis cinerea*. *Phytochemistry*, **57**, 689–692.
- Deighton, N., Muckenschnabel, I., Goodman, B.A. and Williamson, B. (1999) Lipid peroxidation and the oxidative burst associated with infection of *Capsicum annuum* by *Botrytis cinerea*. *Plant J.* **20**, 485–492.
- DeZwaan, T.M., Carroll, A.M., Valent, B. and Sweigard, J.A. (1999) *Magnaporthe grisea* Pth11p is a novel plasma membrane protein that mediates appressorium differentiation in response to inductive substrate cues. *Plant Cell*, **11**, 2013–2030.
- Díaz, J., ten Have, A. and van Kan, J.A.L. (2002) The role of ethylene and wound signaling in resistance of tomato to *Botrytis cinerea*. *Plant Physiol.* **129**, 1341–1351.
- Dickman, M.B., Park, Y.K., Oltersdorf, T., Li, W., Clemente, T. and French, R. (2001) Abrogation of disease development in plants expressing animal anti-apoptotic genes. *Proc. Natl Acad. Sci. USA*, **98**, 6957–6962.
- Dik, A.J. and Wubben, J.P. (2004) Epidemiology of *Botrytis cinerea* diseases in greenhouses. In: *Botrytis: Biology, Pathology and Control*. (Elad, Y., Williamson, B., Tudzynski, P. and Delen, N., eds), pp. 319–333. Dordrecht, The Netherlands: Kluwer Academic Press.
- Diolez, A., Marches, F., Fortini, D. and Brygoo, Y. (1995) Boty, a long-terminal-repeat retroelement in the phytopathogenic fungus *Botrytis cinerea*. *Appl. Env. Microbiol.* **61**, 103–108.
- Dixon, K.P., Xu, J.-R., Smirnov, N. and Talbot, N.J. (1999) Independent signaling pathways regulate cellular turgor during hyperosmotic stress and appressorium-mediated plant infection by *Magnaporthe grisea*. *Plant Cell*, **11**, 2045–2058.
- Döhlemann, G., Berndt, P. and Hahn, M. (2006) Different signalling pathways involving a G-alpha protein, cAMP and a MAP kinase control germination of *Botrytis cinerea* conidia. *Mol. Microbiol.* **59**, 821–835.
- Doss, R.P., Deisenhofer, J., Von Nidda, H.A.K., Soelder, A.H. and McGuire, R.P. (2003) Melanin in the extracellular matrix of germlings of *Botrytis cinerea*. *Phytochemistry*, **63**, 687–691.

- Droby, A. and Lichter, A. (2004) Post-harvest *Botrytis* infection: etiology, development and management. In: *Botrytis: Biology, Pathology and Control*. (Elad, Y., Williamson, B., Tudzynski, P. and Delen, N., eds), pp. 349–367. Dordrecht, The Netherlands: Kluwer Academic Press.
- Elad, Y. and Stewart, A. (2004) Microbial control of *Botrytis* spp. In: *Botrytis: Biology, Pathology and Control*. (Elad, Y., Williamson, B., Tudzynski, P. and Delen, N., eds), pp. 223–241. Dordrecht, The Netherlands: Kluwer Academic Press.
- Elad, Y., Williamson, B., Tudzynski, P. and Delen, N. (2004) *Botrytis: Biology, Pathology and Control*. Dordrecht, The Netherlands: Kluwer Academic Press.
- Engelbrecht, R.W. (2002) *The role of the Mediterranean fruit fly, Ceratitis capitata, in Botrytis bunch rot on grape*. MSc thesis, University of Stellenbosch, South Africa.
- Epton, H.A.S. and Richmond, D.V. (1980) Formation, structure and germination of conidia. In: *The Biology of Botrytis* (Coley-Smith, J.R., Verhooff, K. and Jarvis, W.R., eds), pp. 41–83. London: Academic Press.
- Espino, J.J., Brito, N., Noda, J. and Gonzalez, C. (2005) *Botrytis cinerea* endo- β -1,4-glucanase Cel5A is expressed during infection but is not required for pathogenesis. *Physiol. Mol. Plant Pathol.* **66**, 213–221.
- Faretra, F., Antonacci, E. and Pollastro, S. (1988) Sexual behaviour and mating system of *Botryotinia fuckeliana*, teleomorph of *Botrytis cinerea*. *J. Gen. Microbiol.* **134**, 2543–2550.
- Faretra, F. and Grindle, M. (1992) Genetic studies of *Botryotinia fuckeliana* (*Botrytis cinerea*). In: *Recent Advances in Botrytis Research* (Verhooff, K., Malathrakakis, N.E. and Williamson, B., eds), pp. 7–17. Wageningen, The Netherlands: Pudoc Scientific Publishers.
- Faretra, F. and Pollastro, S. (1991) Genetic basis of resistance to benzimidazole and dicarboximide fungicides in *Botryotinia fuckeliana* (*Botrytis cinerea*). *Mycol. Res.* **95**, 943–951.
- Faretra, F. and Pollastro, S. (1996) Genetic studies of the phytopathogenic fungus *Botryotinia fuckeliana* (*Botrytis cinerea*) by analysis of ordered tetrads. *Mycol. Res.* **100**, 620–624.
- Fermaud, M. and Gaunt, R.E. (1995) *Thrips obscuratus* as a potential vector of *Botrytis cinerea* in kiwifruit. *Mycol. Res.* **99**, 267–273.
- Fermaud, M. and Le Menn, R. (1989) Association of *Botrytis cinerea* with grape berry moth larvae. *Phytopathology*, **79**, 651–656.
- Ferrari, S., Vairo, D., Ausubel, F.M., Cervone, F. and De Lorenzo, G. (2003) Tandemly duplicated *Arabidopsis* genes that encode polygalacturonase-inhibiting proteins are regulated coordinately by different signal transduction pathways in response to fungal infection. *Plant Cell*, **15**, 93–106.
- Finkers, R., van den Berg, P., van Berloo, R., ten Have, A., van Heusden, A.W., van Kan, J.A.L. and Lindhout, P. (2007a) Three QTLs for *Botrytis cinerea* resistance in tomato. *Theor. Appl. Genet.* **114**, 585–593.
- Finkers, R., van Heusden, A.W., Meijer-Dekens, F., van Kan, J.A.L., Maris, P. and Lindhout, P. (2007b) The construction of a *Solanum habrochaites* LYC4 introgression line population and the identification of QTLs for resistance to *Botrytis cinerea*. *Theor. Appl. Genet.* **114**, 1071–1080.
- Fitzgerald, A., van Kan, J.A.L. and Plummer, K.M. (2004) Simultaneous silencing of multiple genes in the apple scab fungus, *Venturia inaequalis*, by expression of RNA with chimeric inverted repeats. *Fungal Genet. Biol.* **41**, 963–971.
- Fournier, E., Giraud, T., Albertini, C. and Brygoo, Y. (2005) Partition of the *Botrytis cinerea* complex in France using multiple gene genealogies. *Mycologia*, **97**, 1251–1267.
- Fournier, E., Giraud, T., Loiseau, A., Vautrin, D., Estoup, A., Solignac, M., Cornuet, J.M. and Brygoo, Y. (2002) Characterization of nine polymorphic microsatellite loci in the fungus *Botrytis cinerea* (Ascomycota). *Mol. Ecol. Notes*, **2**, 253–255.
- Fournier, E., Levis, C., Fortini, D., Leroux, P., Giraud, T. and Brygoo, Y. (2003) Characterization of Bc-hch, the *Botrytis cinerea* homolog of the *Neurospora crassa* het-c vegetative incompatibility locus and its use as a population marker. *Mycologia*, **95**, 251–261.
- Germeier, C., Hedke, K. and von Tiedemann, A. (1994) The use of pH-indicators in diagnostic media for acid-producing plant pathogens. *J. Plant Dis. Protection*, **101**, 498–507.
- Gioti, A., Simon, A., Le Pecheur, P., Giraud, C., Pradier, J.M., Viaud, M. and Levis, C. (2006) Expression profiling of *Botrytis cinerea* genes identifies three patterns of up-regulation in planta and an FKBP12 protein affecting pathogenicity. *J. Mol. Biol.* **358**, 372–386. (Erratum in: *J. Mol. Biol.* (2006). **364**, 550.)
- Giraud, T., Fortini, D., Levis, C., Leroux, P. and Brygoo, Y. (1997) RFLP markers show genetic recombination in *Botryotinia fuckeliana* (*Botrytis cinerea*) and transposable elements reveal two sympatric species. *Mol. Biol. Evol.* **14**, 1177–1185.
- Godoy, G., Steadman, J.R., Dickman, M.B. and Dam, R. (1990) Use of mutants to demonstrate the role of oxalic acid in pathogenicity of *Sclerotinia sclerotiorum* Phaseolus vulgaris. *Physiol. Mol. Plant Pathol.* **36**, 179–191.
- Gossen, B.D. and Platford, G. (1999) Blossom blight in alfalfa seed fields in Saskatchewan and Manitoba 1998. *Can. Plant Dis. Surv.* **79**, 94–95.
- Gourgues, M., Brunet-Simon, A., Lebrun, M.-H. and Levis, C. (2004) The tetraspanin BcPls1 is required for appressorium-mediated penetration of *Botrytis cinerea* into host plant leaves. *Mol. Microbiol.* **51**, 619–629.
- Govrin, E.M. and Levine, A. (2000) The hypersensitive response facilitates plant infection by the necrotrophic pathogen *Botrytis cinerea*. *Curr. Biol.* **10**, 751–757.
- Gubler, W.D., Marois, J.J., Bledsoe, A.M. and Bettiga, L.J. (1987) Control of *Botrytis* bunch rot of grape with canopy management. *Plant Dis.* **71**, 599–601.
- Guimarães, R., Chetelat, R. and Stotz, H. (2004) Resistance to *Botrytis cinerea* in *Solanum lycopersicoides* is dominant in hybrids with tomato, and involves induced hyphal death. *Eur. J. Plant Pathol.* **110**, 13–23.
- Han, Y., Joosten, H.J., Niu, W., Zhao, Z., Mariano, P.S., McCalman, M.T., van Kan, J.A.L., Schaap, P.J. and Dunaway-Mariano, D. (2007) Oxaloacetate hydrolase: the C-C bond lyase of oxalate secreting fungi. *J. Biol. Chem.* **282**, 9581–9590.
- Harrison, J.G. and Lowe, R. (1987) Wind dispersal of conidia of *Botrytis* spp. pathogenic to *Vicia faba*. *Plant Pathol.* **36**, 5–15.
- Harrison, J.G., Lowe, R. and Williams, N.A. (1994) Humidity and fungal diseases of plants — Problems. In: *Ecology of Plant Pathogens* (Blakeman, J.P. and Williamson, B., eds), pp. 79–97. Wallingford: CAB International.
- ten Have, A., Mulder, W., Visser, J. and van Kan, J.A.L. (1998) The endopolygalacturonase gene *Bcpg1* is required for full virulence of *Botrytis cinerea*. *Mol. Plant-Microbe Interact.* **11**, 1009–1016.
- ten Have, A., Oude Breuil, W., Wubben, J.P., Visser, J. and van Kan, J.A.L. (2001) *Botrytis cinerea* endopolygalacturonase genes are differentially expressed in various plant tissues. *Fungal Genet. Biol.* **33**, 97–105.
- ten Have, A., van Berloo, R., Lindhout, P. and van Kan, J.A.L. (2007) Partial stem and leaf resistance against the fungal pathogen *Botrytis cinerea* in wild relatives of tomato. *Eur. J. Plant Pathol.* **117**, 153–166.
- Hayashi, K., Schoonbeek, H.J., Sugiura, H. and De Waard, M.A. (2001) Multidrug resistance in *Botrytis cinerea* with decreased accumulation of

- the azole fungicide oxoconazole and increased transcription of the ABC transporter gene *Bcatrd*. *Pesticide Biochem. Physiol.* **70**, 168–179.
- Hayashi, K., Schoonbeek, H.J., Sugiura, H. and De Waard, M.A. (2002) Expression of the ABC transporter *Bcatrd* from *Botrytis cinerea* reduces sensitivity to sterol demethylation inhibitors. *Pesticide Biochem. Physiol.* **73**, 110–121.
- Hoeberichts, F.A., ten Have, A. and Woltering, E.J. (2003) A tomato metacaspase gene is upregulated during programmed cell death in *Botrytis cinerea*-infected leaves. *Planta*, **217**, 517–522.
- Huang, D., Bhairi, S. and Staples, R.C. (1989) A transformation procedure for *Botrytis squamosa*. *Current Genet.* **15**, 411–414.
- Jarvis, W.R. (1962a) The dispersal of spores of *Botrytis cinerea* Fr. in a raspberry plantation. *Trans. Br. Mycol. Soc.* **45**, 549–559.
- Jarvis, W.R. (1962b) Splash dispersal of spores of *Botrytis cinerea* Pers. *Nature (London)*, **193**, 599.
- Jarvis, W.R. (1980) Taxonomy. In: *The Biology of Botrytis* (Coley-Smith, J.R., Verhoef, K. and Jarvis, W.R., eds), pp. 1–18. London: Academic Press.
- Jenczmionka, N.J. and Schäfer, W. (2005) The Gpmk1 MAP kinase of *Fusarium graminearum* regulates the induction of specific secreted enzymes. *Curr. Genet.* **47**, 29–36.
- Johnson, K.B. and Powelson, M.L. (1983) Analysis of spore dispersal gradients of *Botrytis cinerea* and gray mold disease gradients in snap beans. *Phytopathology*, **73**, 741–746.
- Joubert, A., Kars, I., Wagemakers, C.A.M., Bergman, C., Kemp, G., Vivier, M.A. and van Kan, J.A.L. (2007) A polygalacturonase inhibiting protein from grapevine reduces the symptoms of the endopolygalacturonase BcPG2 from *Botrytis cinerea* in *Nicotiana benthamiana* leaves without any evidence for *in vitro* interaction. *Mol. Plant–Microbe Interact.* **20**, 392–402.
- Joubert, D.A., Slaughter, A.R., Kemp, G., Becker, V.W.J., Krooshof, G.H., Bergmann, C., Benen, J., Pretorius, I.S. and Vivier, M.A. (2006) The grapevine polygalacturonase-inhibiting protein (VvPGIP1) reduces *Botrytis cinerea* susceptibility in transgenic tobacco and differentially inhibits fungal polygalacturonases. *Transgenic Res.* **15**, 687–702.
- Juge, N. (2006) Plant protein inhibitors of cell wall degrading enzymes. *Trends Plant Sci.* **11**, 359–367.
- van Kan, J.A.L. (2006) Licensed to kill: the lifestyle of a necrotrophic plant pathogen. *Trends Plant Sci.* **11**, 247–253.
- van Kan, J.A.L., van't Klooster, J.W., Wagemakers, C.A.M., Dees, D. and van der Vlugt-Bergmans, C.J.B. (1997) Cutinase A of *Botrytis cinerea* is expressed, but not essential, during penetration of gerbera and tomato. *Mol. Plant–Microbe Interact.* **10**, 30–38.
- Kars, I., Krooshof, G., Wagemakers, C.A.M., Joosten, R., Benen, J.A.E. and van Kan, J.A.L. (2005a) Necrotising activity of five *Botrytis cinerea* endopolygalacturonases produced in *Pichia pastoris*. *Plant J.* **43**, 213–225.
- Kars, I., Wagemakers, C.A.M., McCalman, M. and van Kan, J.A.L. (2005b) Functional analysis of *Botrytis cinerea* pectin methylesterase genes by PCR-based targeted mutagenesis: *Bcpme1* and *Bcpme2* are dispensable for virulence of strain B05.10. *Mol. Plant Pathol.* **6**, 641–652.
- Kerssies, A., Bosker-van Zessen, I., Wagemakers, C.A.M. and van Kan, J.A.L. (1997) Variation in pathogenicity and DNA polymorphism among *Botrytis cinerea* isolates sampled inside and outside a glass-house. *Plant Dis.* **81**, 781–786.
- Kliebenstein, D.J., Rowe, H.C. and Denby, K.J. (2005) Secondary metabolites influence Arabidopsis/*Botrytis* interactions: variation in host production and pathogen sensitivity. *Plant J.* **44**, 25–36.
- Klimpel, A., Schulze Gronover, C., Williamson, B., Stewart, J.A. and Tudzynski, B. (2002) The adenylate cyclase (BAC) in *Botrytis cinerea* is required for full pathogenicity. *Mol. Plant Pathol.* **3**, 439–450.
- Kojima, K., Bahn, Y.S. and Heitman, J. (2004) Calcineurin, Mpk1 and Hog1 MAPK pathways independently control fludioxonil antifungal sensitivity in *Cryptococcus neoformans*. *Microbiology*, **152**, 591–604.
- Kovach, J., Petzoldt, R. and Harman, G.E. (2000) Use of honey bees and bumble bees to disseminate *Trichoderma harzianum* 1295-22 to strawberries for *Botrytis* control. *Biol. Control.* **18**, 235–242.
- Kraus, P.R. and Heitman, J. (2003) Coping with stress: calmodulin and calcineurin in model and pathogenic fungi. *Biochem. Biophys. Res. Commun.* **311**, 1151–1157.
- Kronstad, J.W. (1997) Virulence and cAMP in smuts, blights and blights. *Trends Plant Sci.* **2**, 193–199.
- Kulkarni, R.D., Thon, M.R., Pan, H. and Dean, R. (2005) Novel G-protein-coupled receptor-like proteins in the plant pathogenic fungus *Magnaporthe grisea*. *Genome Biol.* **6**, R24.
- Kunz, C., Vandelle, E., Rolland, S., Poinssot, B., Bruel, C., Cimerman, A., Zotti, C., Moreau, E., Vedel, R., Pugin, A. and Boccara, M. (2006) Characterization of a new, nonpathogenic mutant of *Botrytis cinerea* with impaired plant colonization capacity. *New Phytol.* **170**, 537–550.
- Leckie, F., Mattei, B., Capodicasa, C., Hemmings, A., Nuss, L., Aracri, B., De Lorenzo, G. and Cervone, F. (1999) The specificity of polygalacturonase-inhibiting protein (PGIP): a single amino acid substitution in the solvent-exposed beta-strand/beta-turn region of the leucine-rich repeats (LRRs) confers a new recognition capability. *EMBO J.* **18**, 2352–2363.
- Leroux, P. (2004) Chemical control of *Botrytis* and its resistance to chemical fungicides. In: *Botrytis: Biology, Pathology and Control*. (Elad, Y., Williamson, B., Tudzynski, P. and Delen, N., eds), pp. 195–222. Dordrecht, The Netherlands: Kluwer Academic Press.
- Leroux, P., Fritz, R., Debieu, D., Albertini, C., Lanen, C., Bach, J., Gredt, M. and Chapeland, F. (2002) Mechanisms of resistance to fungicides in field strains of *Botrytis cinerea*. *Pest Manage. Sci.* **58**, 876–888.
- Levis, C., Fortini, D. and Brygoo, Y. (1997a) Flipper, a mobile Fot1-like transposable element in *Botrytis cinerea*. *Mol. Gen. Genet.* **254**, 674–680.
- Levis, C., Fortini, D. and Brygoo, Y. (1997b) Transformation of *Botrytis cinerea* with the nitrate reductase gene (*niaD*) shows a high frequency of homologous recombination. *Curr. Genet.* **32**, 157–162.
- Louis, C., Girard, M., Kuhl, G. and Lopez-Ferber, M. (1996) Persistence of *Botrytis cinerea* in its vector *Drosophila melanogaster*. *Phytopathology*, **86**, 934–939.
- Lyon, G.D., Goodman, B.A. and Williamson, B. (2004) *Botrytis cinerea* perturbs redox processes as an attack strategy in plants. In: *Botrytis: Biology, Pathology and Control*. (Elad, Y., Williamson, B., Tudzynski, P. and Delen, N., eds), pp. 119–141. Dordrecht, The Netherlands: Kluwer Academic Press.
- Ma, Z. and Michailides, T.J. (2005) Genetic structure of *Botrytis cinerea* populations from different host plants in California. *Plant Dis.* **10**, 1083–1089.
- Mansfield, J.W. and Richardson, A. (1981) The ultrastructure of interactions between *Botrytis* species and broad bean-leaves. *Physiol. Plant Pathol.* **19**, 41–48.
- Manteau, S., Abouna, S., Lambert, B. and Legendre, L. (2003) Differential regulation by ambient pH of putative virulence factor secretion by the phytopathogenic fungus *Botrytis cinerea*. *FEMS Microbiol. Ecol.* **43**, 359–366.

- Maude, R.B. (1980) Disease control. In: *The Biology of Botrytis* (Coley-Smith, J.R., Verhoef, K. and Jarvis, W.R., eds), pp. 275–308. London: Academic Press.
- McNicol, R.J., Williamson, B. and Dolan, A. (1985) Infection of red raspberry styles and carpels by *Botrytis cinerea* and its possible role in post-harvest grey mould. *Ann. Appl. Biol.* **106**, 49–53.
- Mengiste, T., Chen, X., Salmeron, J. and Dietrich, R. (2003) The BOTRYTIS SUSCEPTIBLE1 gene encodes an R2R3MYB transcription factor protein that is required for biotic and abiotic stress responses in *Arabidopsis*. *Plant Cell*, **15**, 2551–2565.
- Mey, G., Held, K., Scheffer, J., Tenberge, K.B. and Tudzynski, P. (2002) CPMK2, an SLT2-homologous mitogen-activated protein (MAP) kinase, is essential for pathogenesis of *Claviceps purpurea* on rye: evidence for a second conserved pathogenesis-related MAP kinase cascade in phytopathogenic fungi. *Mol. Microbiol.* **46**, 305–318.
- Miura, I., Kamakura, T., Maeno, S., Hayashi, S. and Yamaguchi, I. (1994) Inhibition of enzyme secretion in plant pathogens by mepanipyrim, a novel fungicide. *Pesticide Biochem. Physiol.* **48**, 222–228.
- Moyano, C., Alfonso, C., Gallego, J., Raposo, R. and Melgarejo, P. (2003) Comparison of RAPD and AFLP marker analysis as a means to study the genetic structure of *Botrytis cinerea* populations. *Eur. J. Plant Pathol.* **109**, 515–522.
- Muckenschnabel, I., Goodman, B.A., Deighton, N., Lyon, G.D. and Williamson, B. (2001a) *Botrytis cinerea* induces the formation of free radicals in fruits of *Capsicum annuum* at positions remote from the site of infection. *Protoplasma*, **218**, 112–116.
- Muckenschnabel, I., Goodman, B.A., Williamson, B., Lyon, G.D. and Deighton, N. (2002) Infection of leaves of *Arabidopsis thaliana* by *Botrytis cinerea*: changes in ascorbic acid, free radicals and lipid peroxidation products. *J. Exp. Bot.* **53**, 207–214.
- Muckenschnabel, I., Schulze Gronover, C., Deighton, N., Goodman, B.A., Lyon, G.D., Stewart, D. and Williamson, B. (2003) Oxidative effects in uninfected tissue in leaves of French bean (*Phaseolus vulgaris*) containing soft rots caused by *Botrytis cinerea*. *J. Sci. Food Agric.* **83**, 507–514.
- Muckenschnabel, I., Williamson, B., Goodman, B.A., Lyon, G.D., Stewart, D. and Deighton, N. (2001b) Markers for oxidative stress associated with soft rots in French beans (*Phaseolus vulgaris*) infected by *Botrytis cinerea*. *Planta*, **212**, 376–381.
- Nunes, C., Usall, J., Teixido, N., Torres, R. and Vinas, I. (2002) Control of *Penicillium expansum* and *Botrytis cinerea* on apples and pears with the combination of *Candida sake* and *Pantoea agglomerans*. *J. Food Protection*, **65**, 178–184.
- Pande, S., Singh, G., Rao, J.N., Bakr, M.A., Chaurasia, P.C.P., Joshi, S., Johansen, C., Singh, S.D., Kumar, J., Rahman, M.M. and Gowda, C.L.L. (2002) *Integrated Management of Botrytis Gray Mold of Chickpea*. Information Bulletin no. 61. Patancheru, Andhra Pradesh, India: ICRIASAT.
- Park, S.M., Choi, E.S., Kim, M.J., Cha, B.J., Yang, M.S. and Kim, D.H. (2004) Characterization of HOG1 homologue, Cpmk1, from *Cryphonectria parasitica* and evidence for hypovirus-mediated perturbation of its phosphorylation in response to hypertonic stress. *Mol. Microbiol.* **51**, 1267–1277.
- Peng, G., Sutton, J.C. and Kevan, P.G. (1992) Effectiveness of honey bees for applying the biocontrol agent *Gliocladium roseum* to strawberry flowers to suppress *Botrytis cinerea*. *Can. J. Plant Pathol.* **14**, 117–129.
- Pollastro, S., Faretra, F., Di Canio, V. and De Guido, A. (1996) Characterization and genetic analysis of field isolates of *Botryotinia fuckeliana* (*Botrytis cinerea*) resistant to dichlofluanid. *Eur. J. Plant Pathol.* **102**, 607–613.
- Powell, A.L.T., van Kan, J.A.L., ten Have, A., Visser, J., Greve, L.C., Bennett, A.B. and Labavitch, J.M. (2000) Transgenic expression of pear PGIP in tomato limits fungal colonization. *Mol. Plant–Microbe Interact.* **13**, 942–950.
- Quidde, T., Buttner, P. and Tudzynski, P. (1999) Evidence for three different specific saponin-detoxifying activities in *Botrytis cinerea* and cloning and functional analysis of a gene coding for a putative avenacinase. *Eur. J. Plant Pathol.* **105**, 273–283.
- Raposo, R., Gomez, V., Urrutia, T. and Melgarejo, P. (2000) Fitness of *Botrytis cinerea* associated with dicarboximide resistance. *Phytopathology*, **90**, 1246–1249.
- Reino, J.L., Hernández-Galán, R., Durán-Patrón, R. and Collado, I.G. (2004) Virulence-toxin production relationship in isolates of the plant pathogenic fungus *Botrytis cinerea*. *J. Phytopathol.* **152**, 563–566.
- Reis, H., Pfiffli, S. and Hahn, M. (2005) Molecular and functional characterization of a secreted lipase from *Botrytis cinerea*. *Mol. Plant Pathol.* **6**, 257–267.
- Reuveni, R. and Raviv, M. (1992) The effect of spectrally modified polyethylene films on the development of *Botrytis cinerea* in greenhouse grown tomato plants. *Biol. Agric. Horticulture*, **9**, 77–86.
- Reuveni, R., Raviv, M. and Bar, R. (1989) Sporulation of *Botrytis cinerea* as affected by photosensitive sheets and filters. *Ann. Appl. Biol.* **115**, 417–424.
- Rolke, Y., Liu, S., Quidde, T., Williamson, B., Schouten, S., Weltring, K.-M., Siewers, V., Tenberge, K.B., Tudzynski, B. and Tudzynski, P. (2004) Functional analysis of H₂O₂-generating systems in *Botrytis cinerea*: the major Cu-Zn-superoxide dismutase (BCSD1) contributes to virulence on French bean, whereas a glucose oxidase (BCGOD1) is dispensable. *Mol. Plant Pathol.* **5**, 17–27.
- Rolland, S., Jobic, C., Fevre, M. and Bruel, C. (2003) *Agrobacterium*-mediated transformation of *Botrytis cinerea*, simple purification of monokaryotic transformants and rapid conidia-based identification of the transfer-DNA host genomic DNA flanking sequences. *Current Genet.* **44**, 164–171.
- Rosslenbroich, H.-J. and Stuebler, D. (2000) *Botrytis cinerea*—history of chemical control and novel fungicides for its management. *Crop Protection*, **19**, 557–561.
- Rui, O. and Hahn, M. (2007) The SlT2-type Map kinase Bmp3 of *Botrytis cinerea* is required for normal saprotrophic growth, conidiation, plant surface sensing, and host tissue colonization. *Mol. Plant Pathol.* **8**, 173–184.
- Salinas, J., Glandorf, D.C.M., Picavet, F.D. and Verhoef, K. (1989) Effects of temperature, relative humidity and age of conidia on the incidence of spotting on gerbera flowers caused by *Botrytis cinerea*. *Neth. J. Plant Pathol.* **95**, 51–64.
- Schoonbeek, H., Del Sorbo, G. and De Waard, M.A. (2001) The ABC transporter BcatrB affects the sensitivity of *Botrytis cinerea* to the phytoalexin resveratrol and the fungicide fenpiclonil. *Mol. Plant–Microbe Interact.* **14**, 562–571.
- Schouten, A., Tenberge, K.B., Vermeer, J., Stewart, J., Wagemakers, C.A.M., Williamson, B. and van Kan, J.A.L. (2002) Functional analysis of an extracellular catalase of *Botrytis cinerea*. *Mol. Plant Pathol.* **3**, 227–238.
- Schulze Gronover, C., Kasulke, D., Tudzynski, P. and Tudzynski, B. (2001) The role of G protein alpha subunits in the infection process of the gray mold fungus *Botrytis cinerea*. *Mol. Plant–Microbe Interact.* **14**, 1293–1302.
- Schulze Gronover, C., Schorn, C. and Tudzynski, B. (2004) Identification of *Botrytis cinerea* genes up-regulated during infection and controlled by

- the Ga subunit BCG1 using suppression subtractive hybridization (SSH). *Mol. Plant-Microbe Interact.* **17**, 537–546.
- Schulze Gronover, C., Schumacher, J., Hantsch, P. and Tudzynski, B. (2005) A novel seven-helix transmembrane protein BTP1 of *Botrytis cinerea* controls expression of GST-encoding genes, but is not essential for pathogenicity. *Mol. Plant Pathol.* **6**, 243–256.
- Segmüller, N., Ellendorf, U., Tudzynski, B. and Tudzynski, P. (2007) BcSAK1, a stress-activated MAP kinase is involved in vegetative differentiation and pathogenicity in *Botrytis cinerea*. *Eukaryot. Cell*, **6**, 211–221.
- Shirane, N., Masuko, M. and Hayashi, Y. (1988) Nuclear behaviour and division in germinating conidia of *Botrytis cinerea*. *Phytopathology*, **78**, 1627–1630.
- Siewers, V., Viaud, M., Jimenez-Teja, D., Collado, I.G., Schulze Gronover, C., Pradier, J.-M., Tudzynski, B. and Tudzynski, P. (2005) Functional analysis of the cytochrome P450 monooxygenase gene *Bcbot1* of *Botrytis cinerea* indicates that botrydial is a strain-specific virulence factor. *Mol. Plant-Microbe Interact.* **18**, 602–612.
- Siriputthaiwan, P., Jauneau, A., Herbert, C., Garcin, D. and Dumas, B. (2005) Functional analysis of CLPT1, a Rab/GTPase required for protein secretion and pathogenesis in the plant fungal pathogen *Colletotrichum lindemuthianum*. *J. Cell Sci.* **118**, 323–329.
- Solomon, P.S., Waters, O.D., Simmonds, J., Cooper, R.M. and Oliver, R. (2005) The Mak2 MAP kinase signal transduction pathway is required for pathogenicity in *Stagonospora nodorum*. *Curr. Genet.* **48**, 60–68.
- Staats, M., van Baarlen, P., Schouten, A., van Kan, J.A.L. and Bakker, F.T. (2007) Positive selection in phytotoxic protein-encoding genes of *Botrytis* species. *Fungal Genet. Biol.* **44**, 52–63.
- Staats, M., van Baarlen, P. and van Kan, J.A.L. (2005) Molecular phylogeny of the plant pathogenic genus *Botrytis* and the evolution of host specificity. *Mol. Biol. Evol.* **22**, 333–346.
- Tenberge, K.B. (2004) Morphology and cellular organization in *Botrytis* interactions with plants. In: *Botrytis: Biology, Pathology and Control*. (Elad, Y., Williamson, B., Tudzynski, P., and and Delen, N., eds), pp. 67–84. Dordrecht, The Netherlands: Kluwer Academic Publishers.
- Thomma, B.P., Eggermont, K., Tierens, K.F. and Broekaert, W.F. (1999) Requirement of functional ethylene-insensitive 2 gene for efficient resistance of Arabidopsis to infection by *Botrytis cinerea*. *Plant Physiol.* **121**, 1093–1102.
- Thomma, B.P., Nelissen, I., Eggermont, K. and Broekaert, W.F. (1998) Deficiency in phytoalexin production causes enhanced susceptibility of *Arabidopsis thaliana* to the fungus *Alternaria brassicicola*. *Plant J.* **19**, 163–171.
- Thomma, B.P., Penninckx, I.A., Cammue, B.P. and Broekaert, W.F. (2001) The complexity of disease signaling in Arabidopsis. *Curr. Opin. Immunol.* **13**, 63–68.
- Tudzynski, B. and Schulze Gronover, C. (2004) Signalling in *Botrytis cinerea*. In: *Botrytis: Biology, Pathology and Control*. (Elad, Y., Williamson, B., Tudzynski, P. and Delen, N., eds), pp. 85–98. Dordrecht, The Netherlands: Kluwer Academic Press.
- Tudzynski, P. and Siewers, V. (2004) Approaches to molecular genetics and genomics of *Botrytis*. In: *Botrytis: Biology, Pathology and Control*. (Elad, Y., Williamson, B., Tudzynski, P. and Delen, N., eds), pp. 53–66. Dordrecht, The Netherlands: Kluwer Academic Press.
- Valette-Collet, O., Cimerman, A., Reignault, P., Levis, C. and Boccara, M. (2003) Disruption of *Botrytis cinerea* pectin methylesterase gene *Bcpme1* reduces virulence on several host plants. *Mol. Plant-Microbe Interact.* **16**, 360–367.
- Verhoeff, K., Leeman, M., van Peer, R., Posthuma, L., Schot, N. and van Eijk, G.W. (1988) Changes in pH and the production of organic acids during colonization of tomato petioles by *Botrytis cinerea*. *J. Phytopathol.* **122**, 327–336.
- Viaud, M., Brunet-Simon, A., Brygoo, Y., Pradier, J.-M. and Levis, C. (2003) Cyclophilin A and calcineurin functions investigated by gene inactivation, cyclosporin A inhibition and cDNA arrays approaches in the phytopathogenic fungus *Botrytis cinerea*. *Mol. Microbiol.* **50**, 1451–1465.
- Viaud, M., Fillinger, S., Liu, W., Polepalli, J.S., Le Pecqueur, P., Kunduru, A.R., Leroux, P. and Legendre, L. (2006) A class III histidine kinase acts as a novel virulence factor in *Botrytis cinerea*. *Mol. Plant-Microbe Interact.* **19**, 1042–1050.
- Viret, O., Keller, M., Jaudzems, V.G. and Cole, M.F. (2004) *Botrytis cinerea* infection of grape flowers: light and electron microscopical studies of infection sites. *Phytopathology*, **94**, 850–857.
- van der Vlugt-Bergmans, C.J.B., Brandwagt, B.F., van't Klooster, J.W., Wagemakers, C.A.M. and van Kan, J.A.L. (1993) Genetic variation and segregation of DNA polymorphisms in *Botrytis cinerea*. *Mycol. Res.* **97**, 1193–1200.
- de Waard, M.A., Andrade, A.C., Hayashi, K., Schoonbeek, H., Stergiopoulos, I. and Zwiers, L.-H. (2006) Impact of fungal drug transporters on fungicide sensitivity, multidrug resistance and virulence. *Pest Manage. Sci.* **62**, 195–207.
- West, J.S., Pearson, S., Hadley, P., Wheldon, A.E., Davis, F.J., Gilbert, A. and Henbest, R.G.C. (2000) Spectral filters for the control of *Botrytis cinerea*. *Ann. Appl. Biol.* **136**, 115–120.
- Williamson, B., Duncan, G.H., Harrison, J.G., Harding, L.A., Elad, Y. and Zimand, G. (1995) Effect of humidity on infection of rose petals by dry-inoculated conidia of *Botrytis cinerea*. *Mycol. Res.* **99**, 1303–1310.
- Williamson, B., McNicol, R.J. and Dolan, A. (1987) The effect of inoculating flowers and developing fruits with *Botrytis cinerea* on post-harvest grey mould of red raspberry. *Ann. Appl. Biol.* **111**, 285–294.
- Wolanin, P.M., Thomason, P.A. and Stock, J.B. (2002) Histidine protein kinases: key signal transducers outside the animal kingdom. *Genome Biol.* **3**, Review 3013.7.
- Woodford, J.A.T., Williamson, B. and Gordon, S.C. (2002) Raspberry beetle damage decreases shelf-life of raspberries also infected with *Botrytis cinerea*. *Acta Hort.* **585**, 423–426.
- Wubben, J.P., Mulder, W., ten Have, A., van Kan, J.A.L. and Visser, J. (1999) Cloning and partial characterisation of endopolygalacturonase genes from *Botrytis cinerea*. *Appl. Environ. Microbiol.* **65**, 1596–1602.
- Xiao, C.L., Chandler, C.K., Price, J.F., Duval, J.R., Mertely, J.C. and Legard, D.E. (2001) Comparison of epidemics of Botrytis fruit rot and powdery mildew of strawberry in large plastic tunnel and field production systems. *Plant Dis.* **85**, 901–909.
- Xu, J.-R. (2000) MAP kinases in fungal pathogens. *Fungal Genet. Biol.* **31**, 137–152.
- Xu, J.-R. and Hamer, J.E. (1996) MAP kinase and cAMP signaling regulate infection structure formation and pathogenic growth in the rice blast fungus *Magnaporthe grisea*. *Genes Dev.* **10**, 2696–2706.
- Zheng, L., Campbell, M., Murphy, J., Lam, S. and Xu, J.-R. (2000) The *BMP1* gene is essential for pathogenicity in the grey mold fungus *Botrytis cinerea*. *Mol. Plant-Microbe Interact.* **13**, 724–732.